>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A TRUSS ELEMENT: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

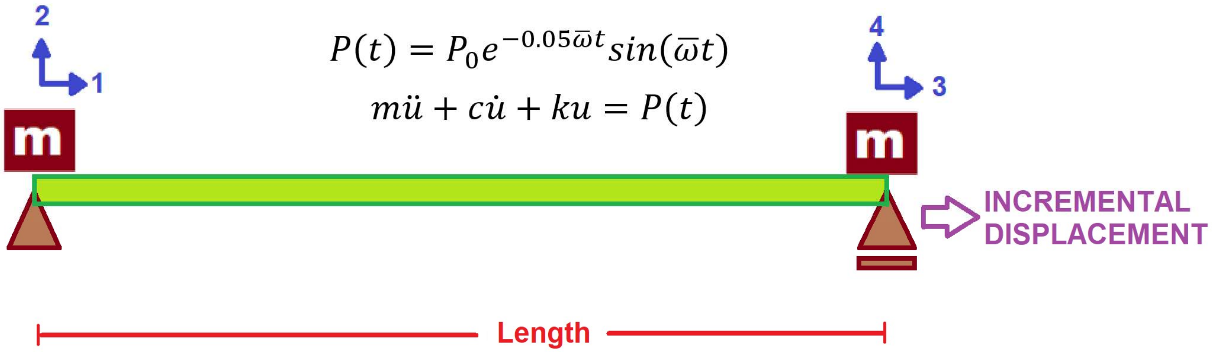
THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

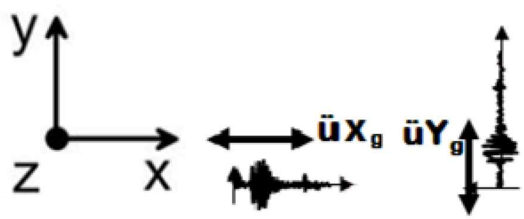
$$Axial\ Ductility\ Damage\ Index = \frac{\varepsilon_d - \varepsilon_y}{\varepsilon_u - \varepsilon_y}$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

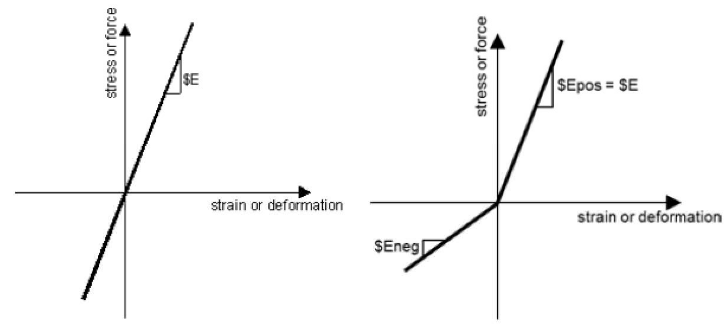
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

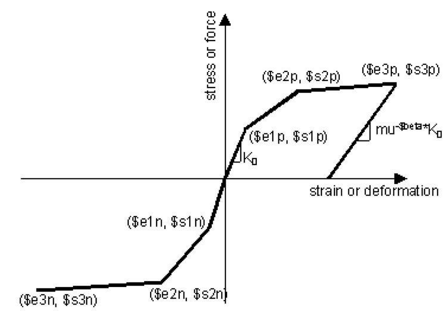




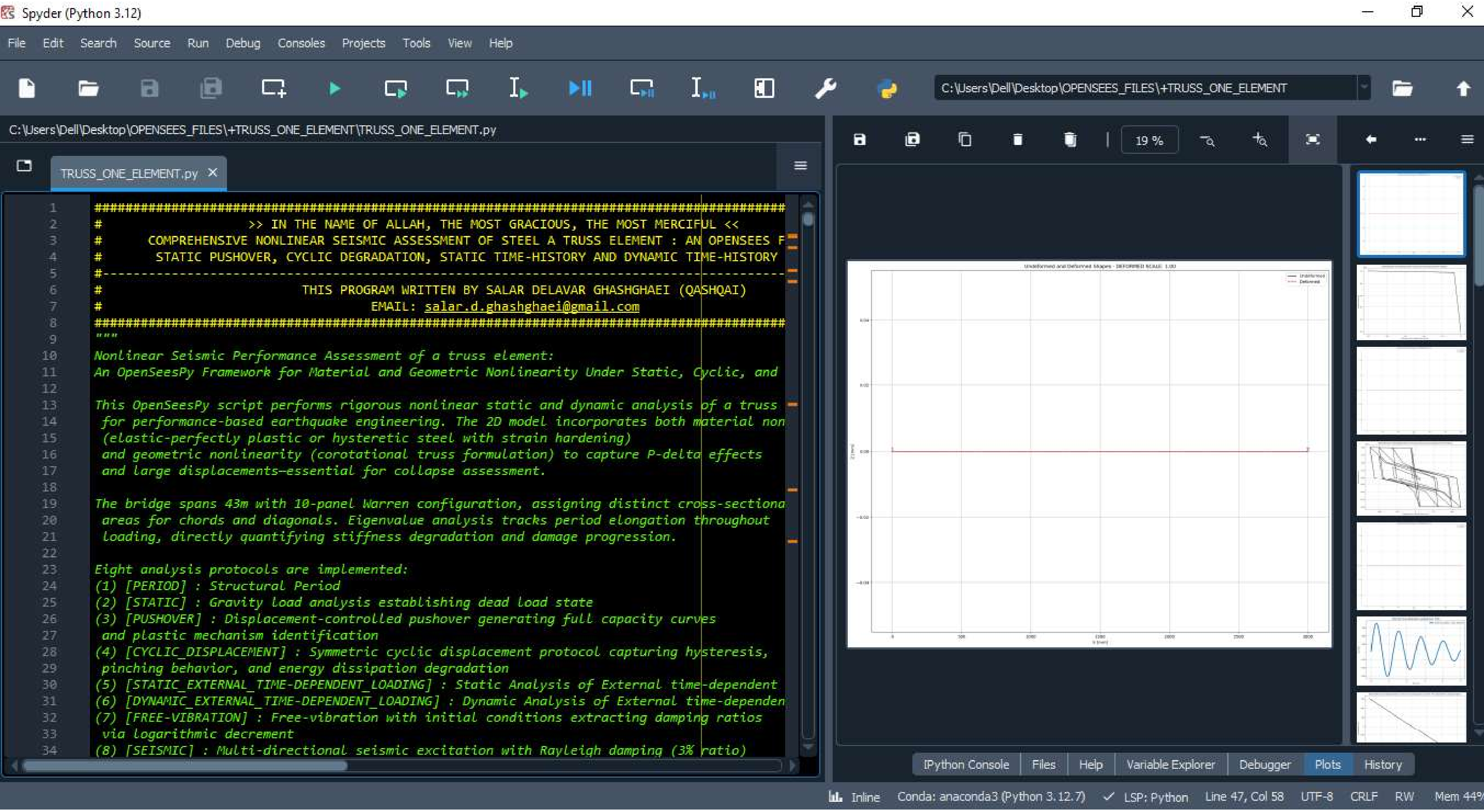
$$\rho A \frac{\partial^2 v}{\partial t^2} = P(t)$$



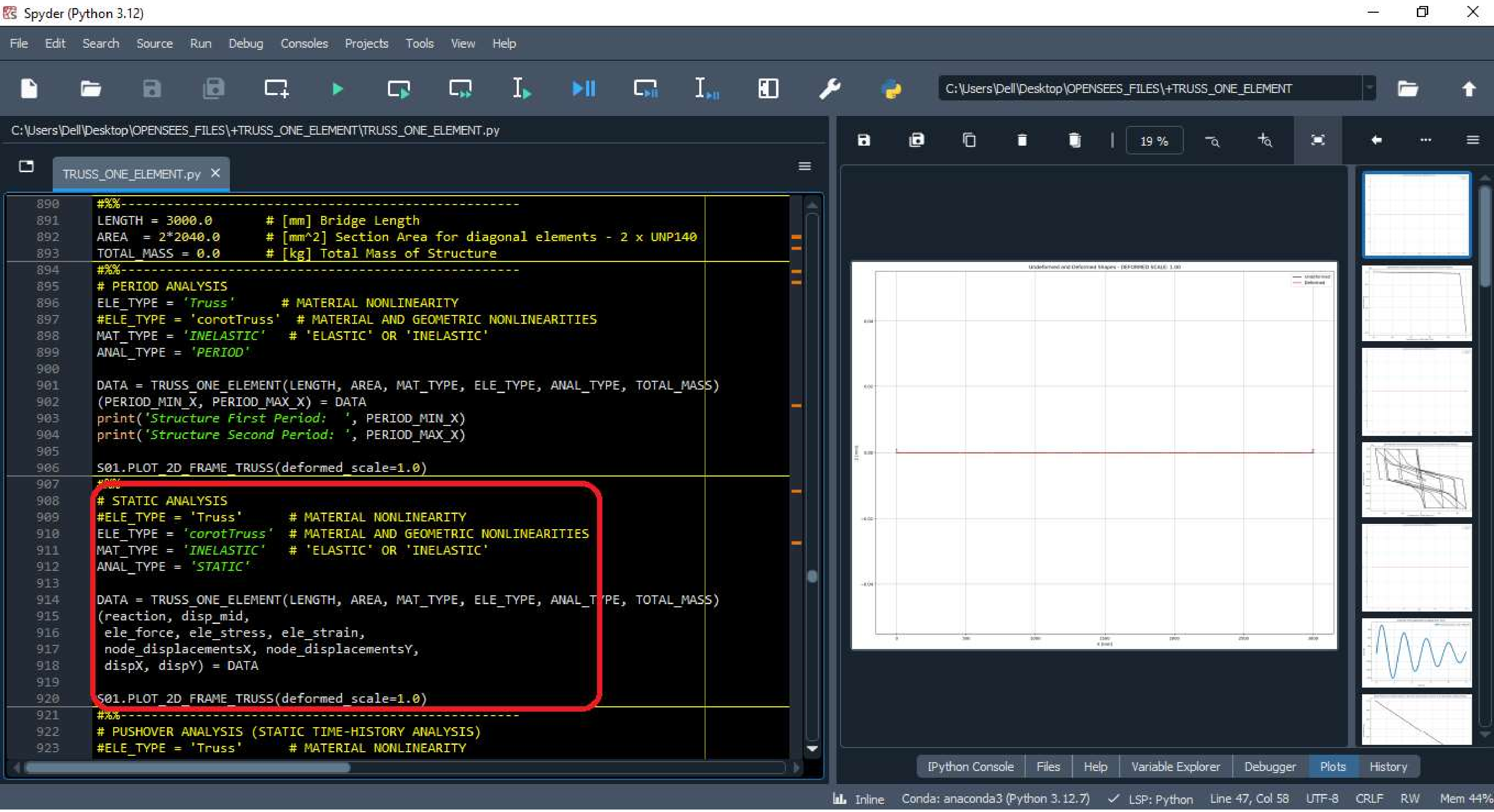
TRUSS ELEMENT ELASTIC MATERIAL



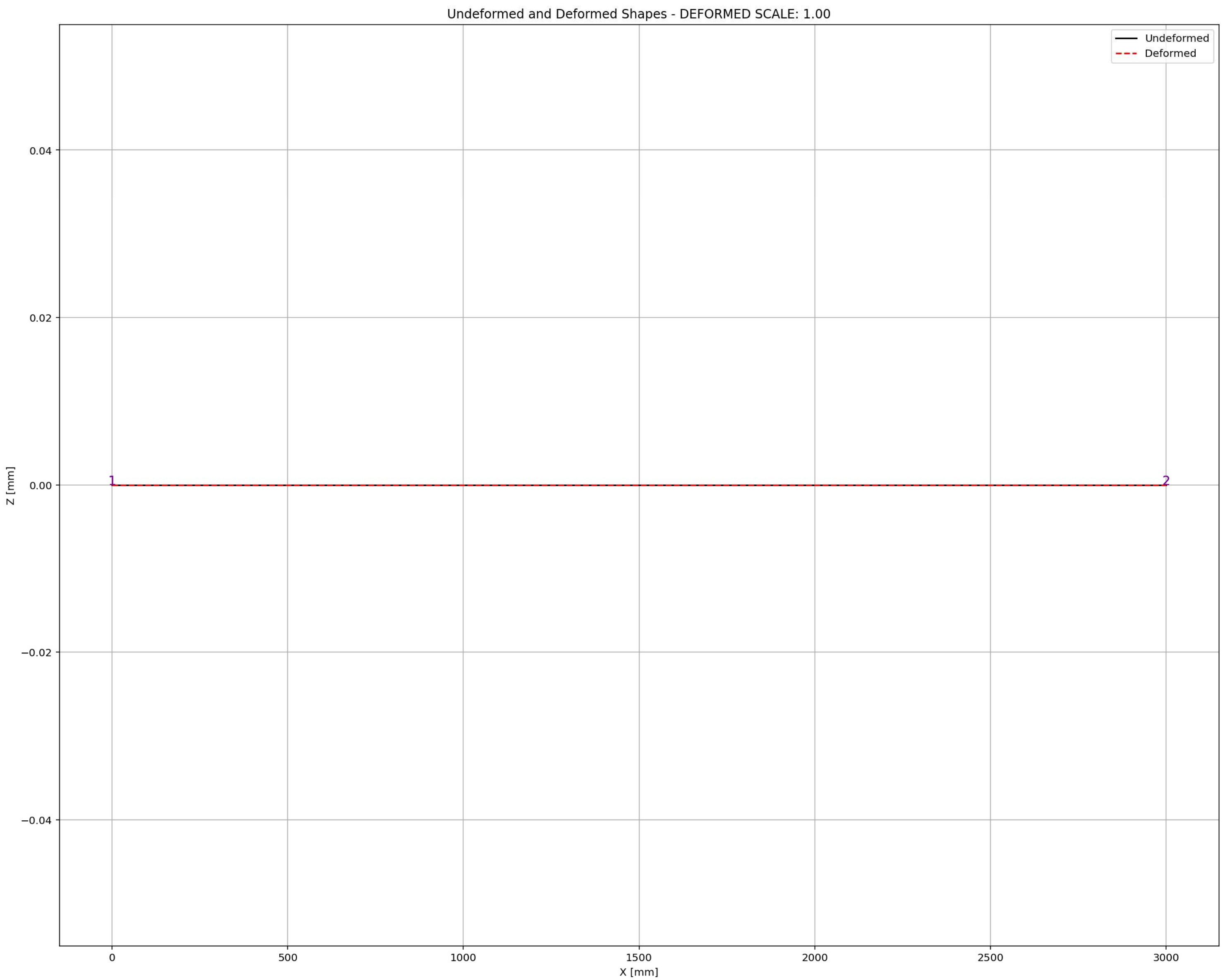
TRUSS ELEMENT INELASTIC MATERIAL


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A TRUSS ELEMENT : AN OPENSEES F
STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
Nonlinear Seismic Performance Assessment of a truss element:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a truss
for performance-based earthquake engineering. The 2D model incorporates both material non
(elastic-perfectly plastic or hysteretic steel with strain hardening)
and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
and large displacements—essential for collapse assessment.
The bridge spans 43m with 10-panel Warren configuration, assigning distinct cross-sectiona
areas for chords and diagonals. Eigenvalue analysis tracks period elongation throughout
loading, directly quantifying stiffness degradation and damage progression.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
and plastic mechanism identification
(4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
pinching behavior, and energy dissipation degradation
(5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
(6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
(7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
via logarithmic decrement
(8) [SEISMIC] : Multi-directional seismic excitation with Rayleigh damping (3% ratio)
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 44%

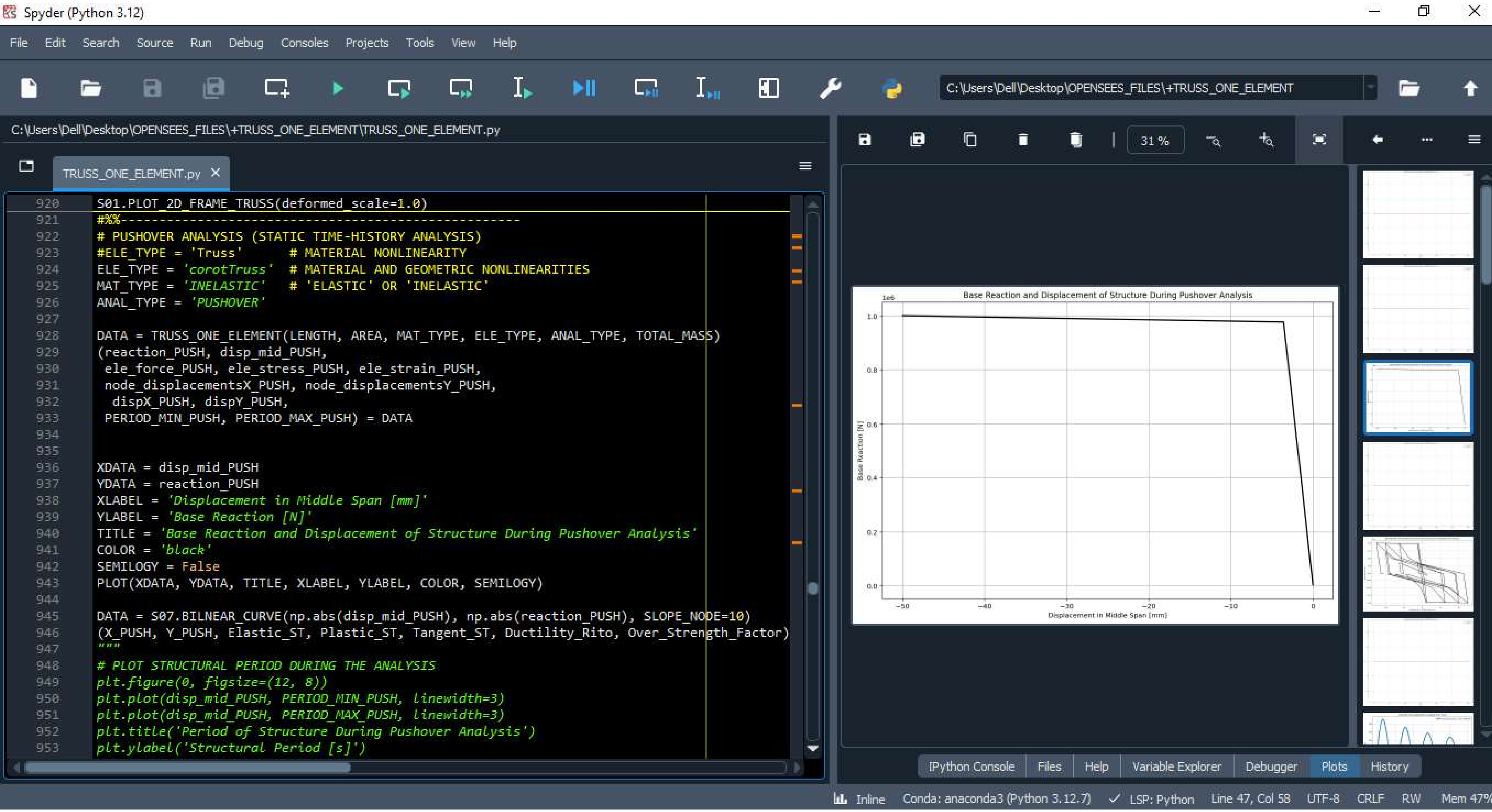
STATIC ANALYSIS



Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
#%%--
LENGTH = 3000.0   # [mm] Bridge Length
AREA = 2*2040.0   # [mm^2] Section Area for diagonal elements - 2 x UNP140
TOTAL_MASS = 0.0   # [kg] Total Mass of Structure
#%%--
# PERIOD ANALYSIS
ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
#ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PERIOD'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(PERIOD_MIN_X, PERIOD_MAX_X) = DATA
print('Structure First Period: ', PERIOD_MIN_X)
print('Structure Second Period: ', PERIOD_MAX_X)
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
#%%
STATIC ANALYSIS
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction, disp_mid,
 ele_force, ele_stress, ele_strain,
 node_displacementsX, node_displacementsY,
 dispX, dispY) = DATA
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
#%%--
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 44%


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.04
0.02
0.00
−0.02
−0.04
0
500
1000
1500
2000
2500
3000
1
2

PUSHOVER ANALYSIS



Spyder (Python 3.12)
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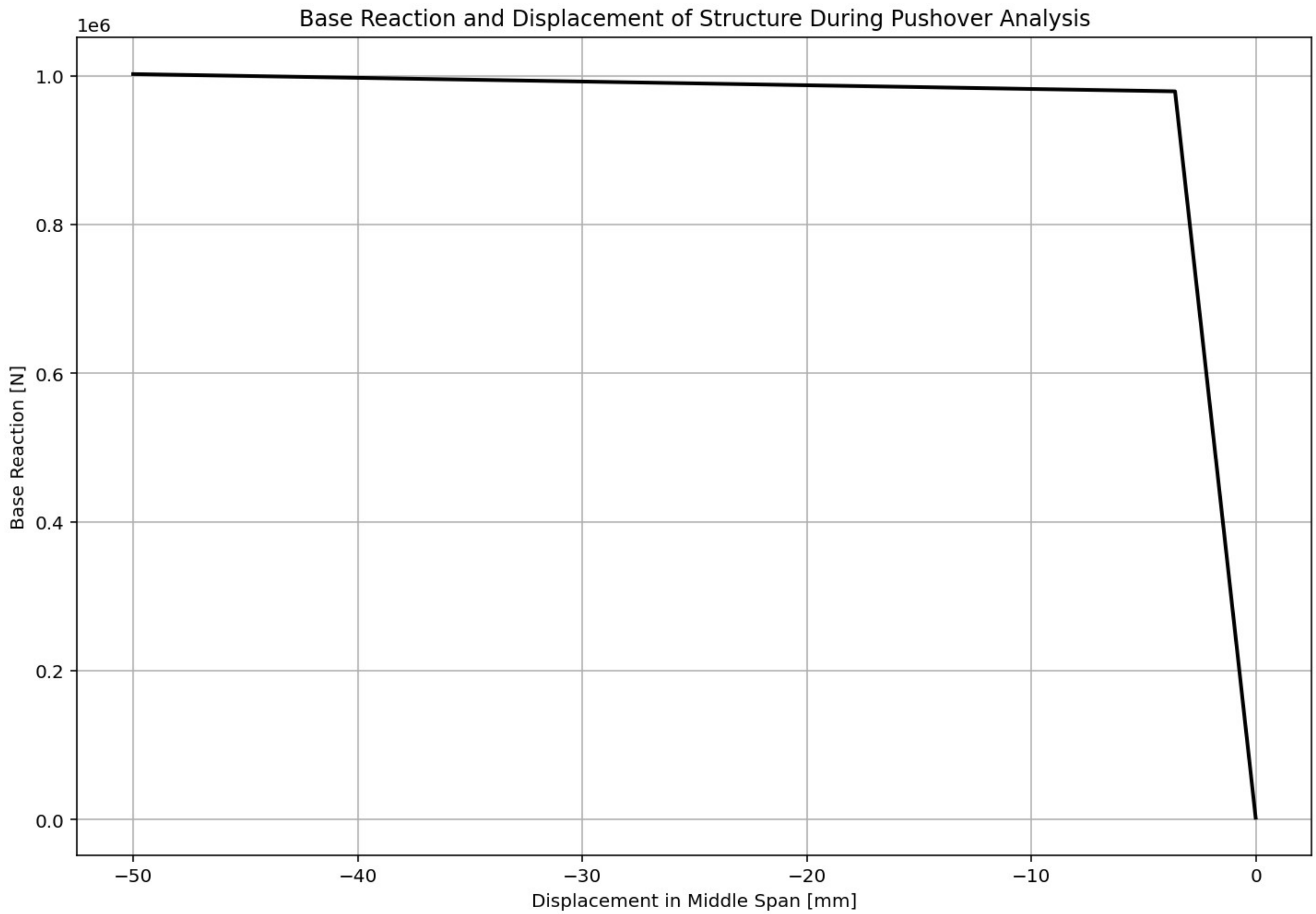
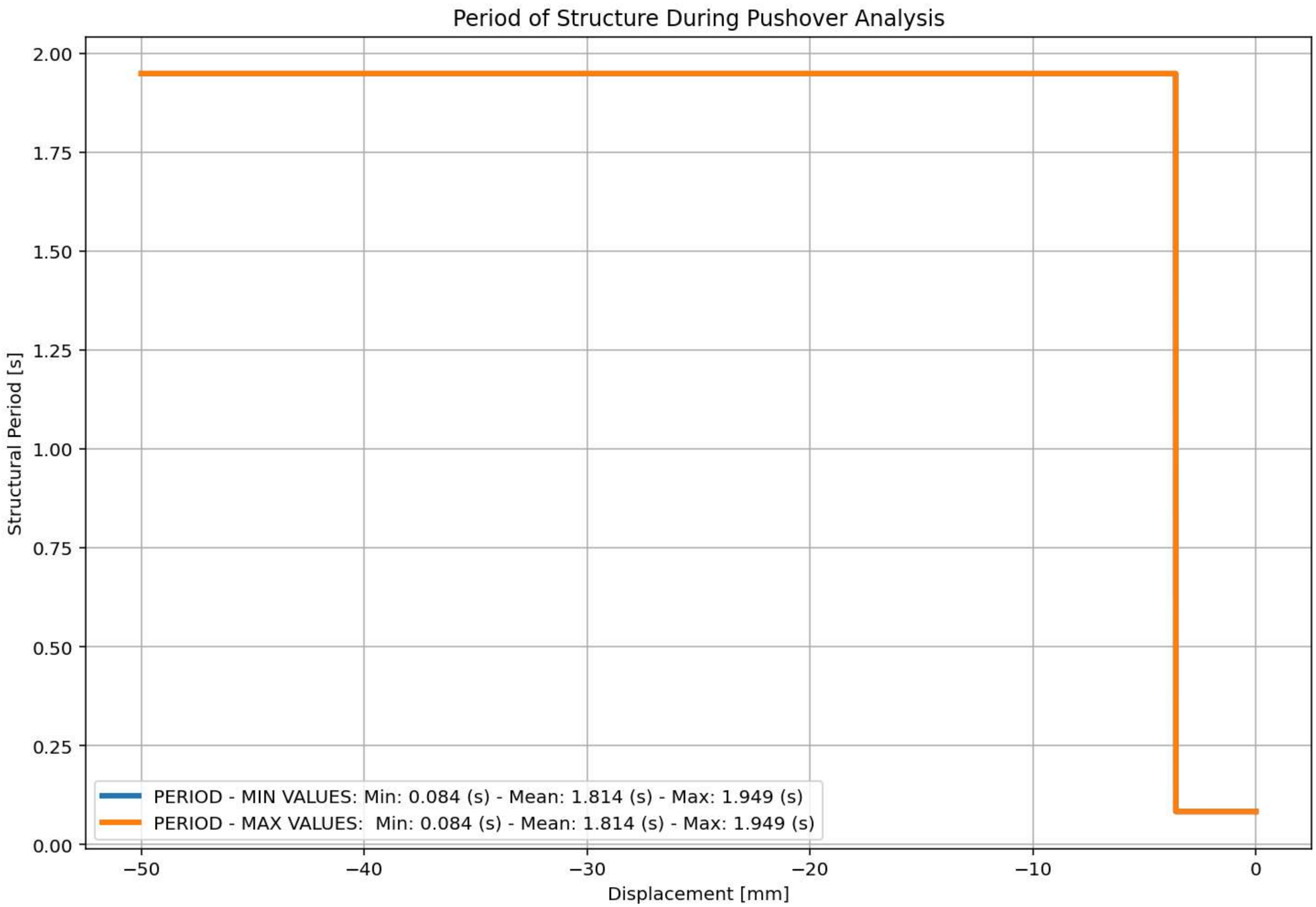
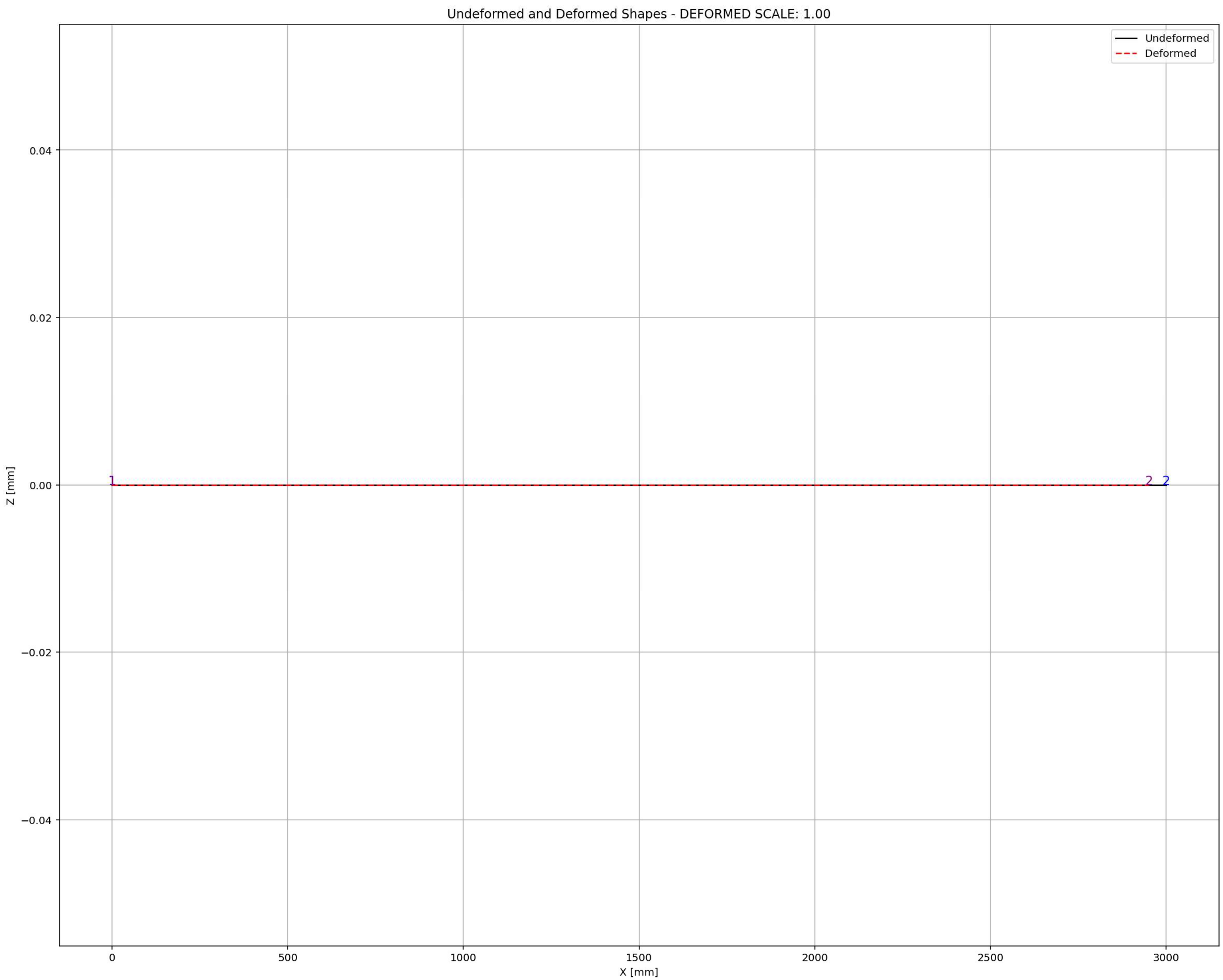
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
#%%--
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction_PUSH, disp_mid_PUSH,
 ele_force_PUSH, ele_stress_PUSH, ele_strain_PUSH,
 node_displacementsX_PUSH, node_displacementsY_PUSH,
  dispX_PUSH, dispY_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
XDATA = disp_mid_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
(X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor)
"""
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
plt.title('Period of Structure During Pushover Analysis')
plt.ylabel('Structural Period [s]')
31 %
Base Reaction and Displacement of Structure During Pushover Analysis
1e6
1.0
0.8
0.6
0.4
0.2
0.0
Base Reaction [N]
−50
−40
−30
−20
−10
0
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 47%

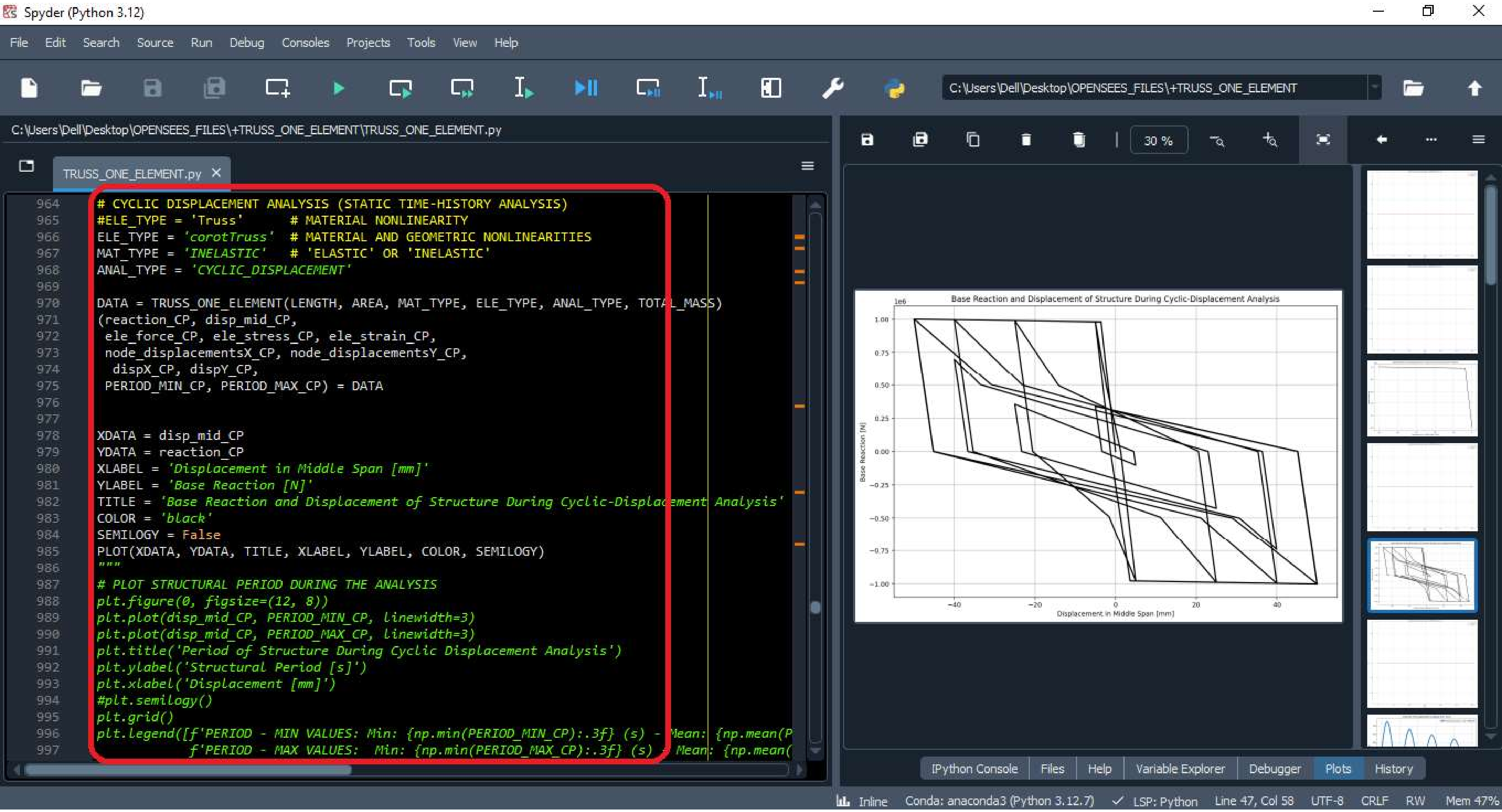

Base Reaction and Displacement of Structure During Pushover Analysis
1e6
1.0
0.8
0.6
0.4
0.2
0.0
Base Reaction [N]
−50
−40
−30
−20
−10
0
Displacement in Middle Span [mm]


Period of Structure During Pushover Analysis
Structural Period [s]
Displacement [mm]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−50
−40
−30
−20
−10
0
PERIOD - MIN VALUES: Min: 0.084 (s) - Mean: 1.814 (s) - Max: 1.949 (s)
PERIOD - MAX VALUES: Min: 0.084 (s) - Mean: 1.814 (s) - Max: 1.949 (s)



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
0.04
0.02
0.00
−0.02
−0.04
Z [mm]
0
500
1000
1500
2000
2500
3000
X [mm]
1
2
2

CYCLIC DISPLACEMENT ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
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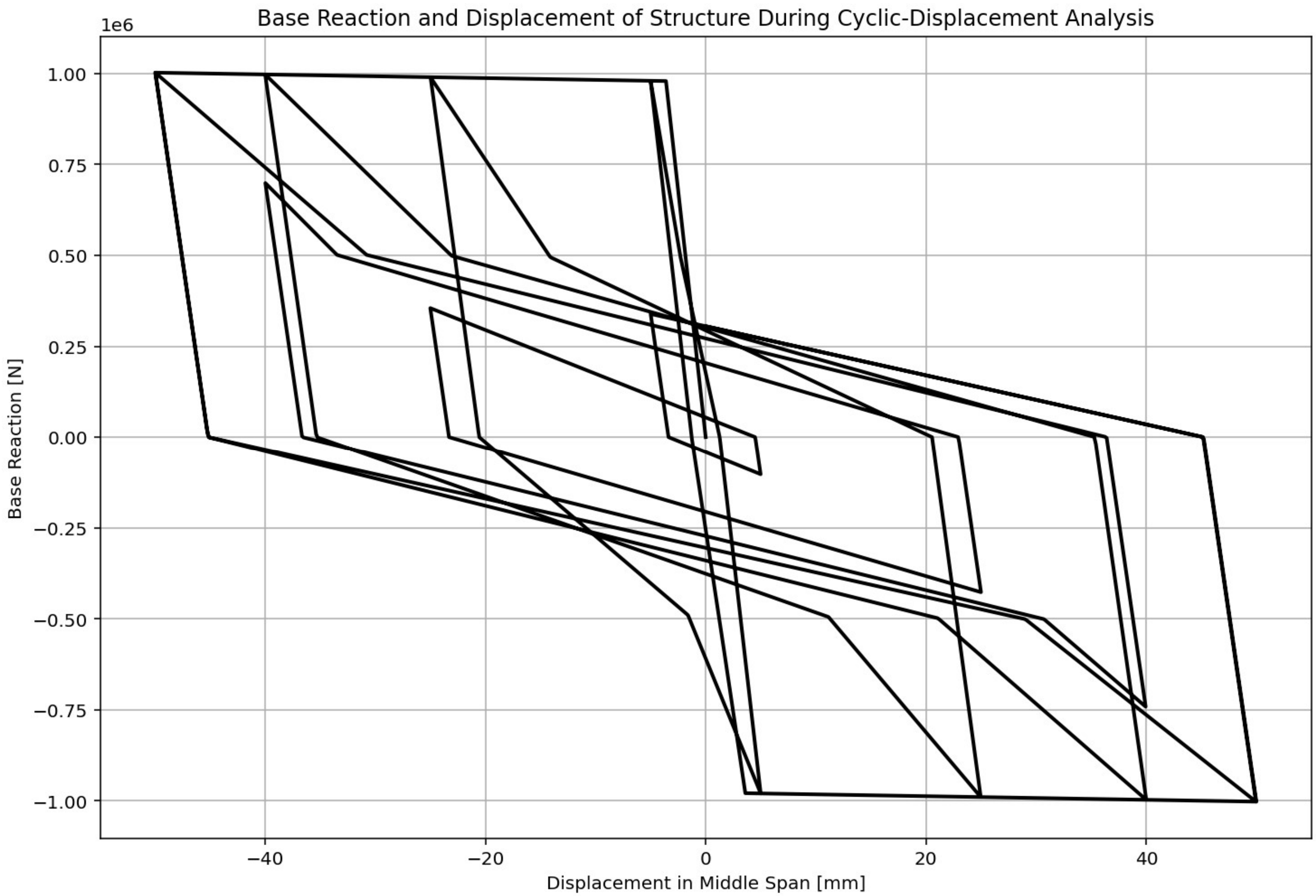
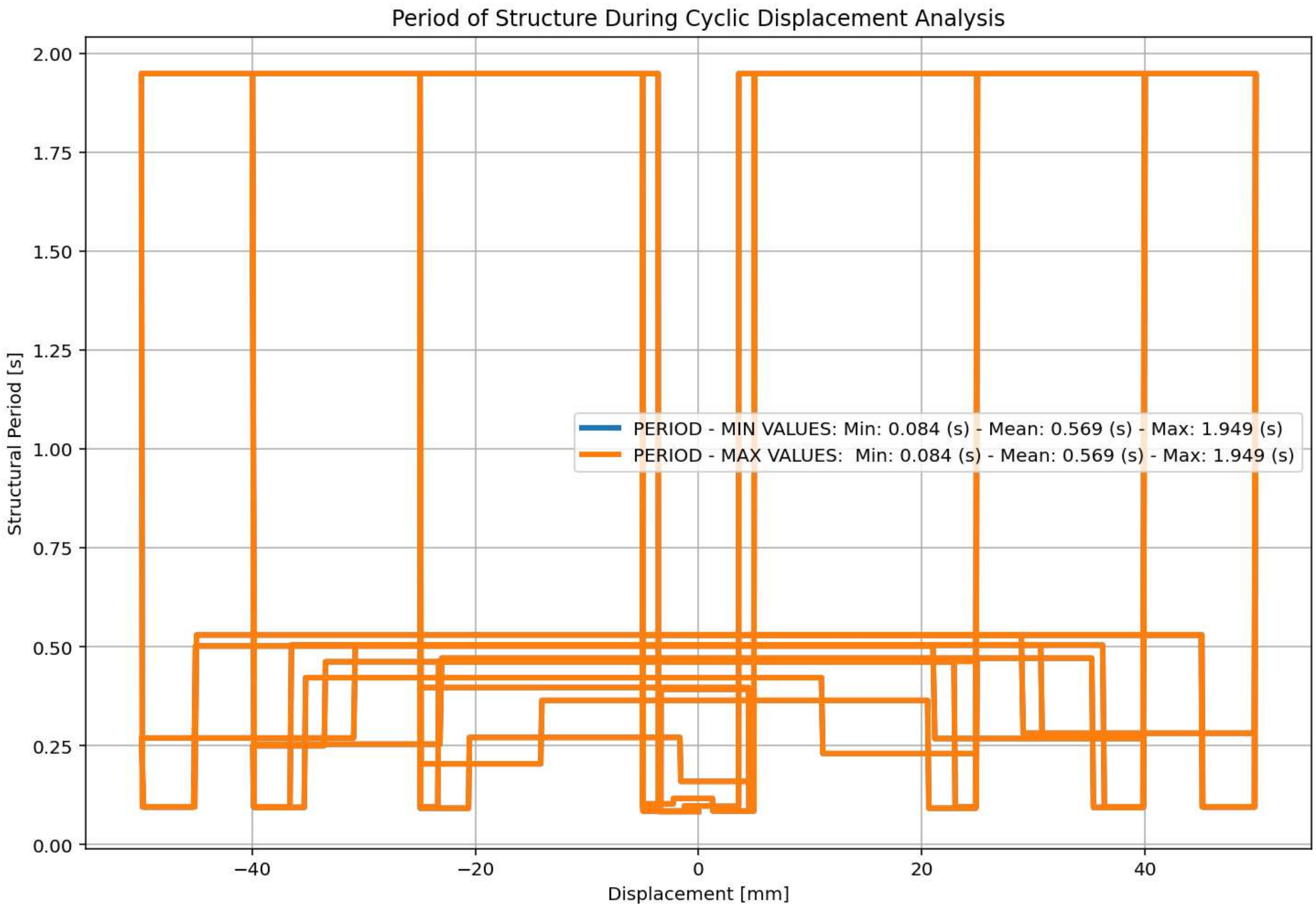
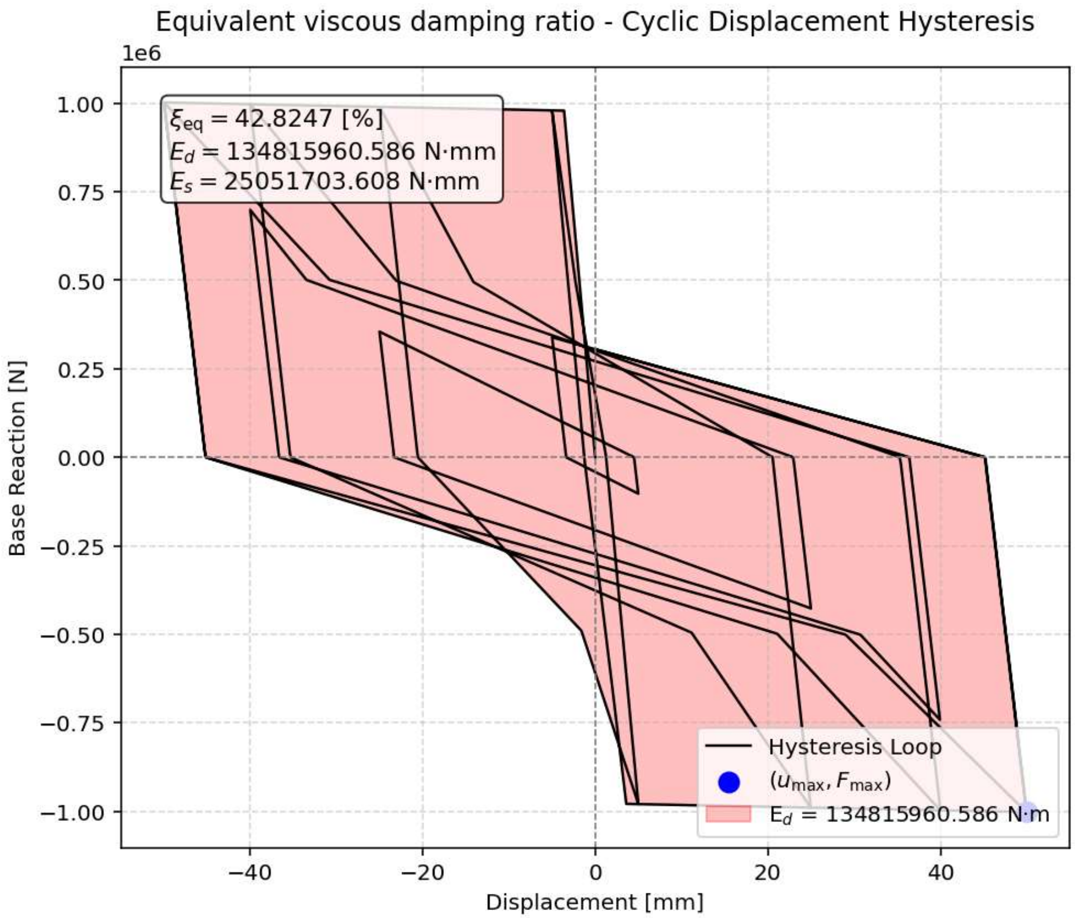
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction_CP, disp_mid_CP,
 ele_force_CP, ele_stress_CP, ele_strain_CP,
 node_displacementsX_CP, node_displacementsY_CP,
  dispX_CP, dispY_CP,
 PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
XDATA = disp_mid_CP
YDATA = reaction_CP
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
"""
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
plt.title('Period of Structure During Cyclic Displacement Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(P
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(
30 %
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
1e6
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 47%

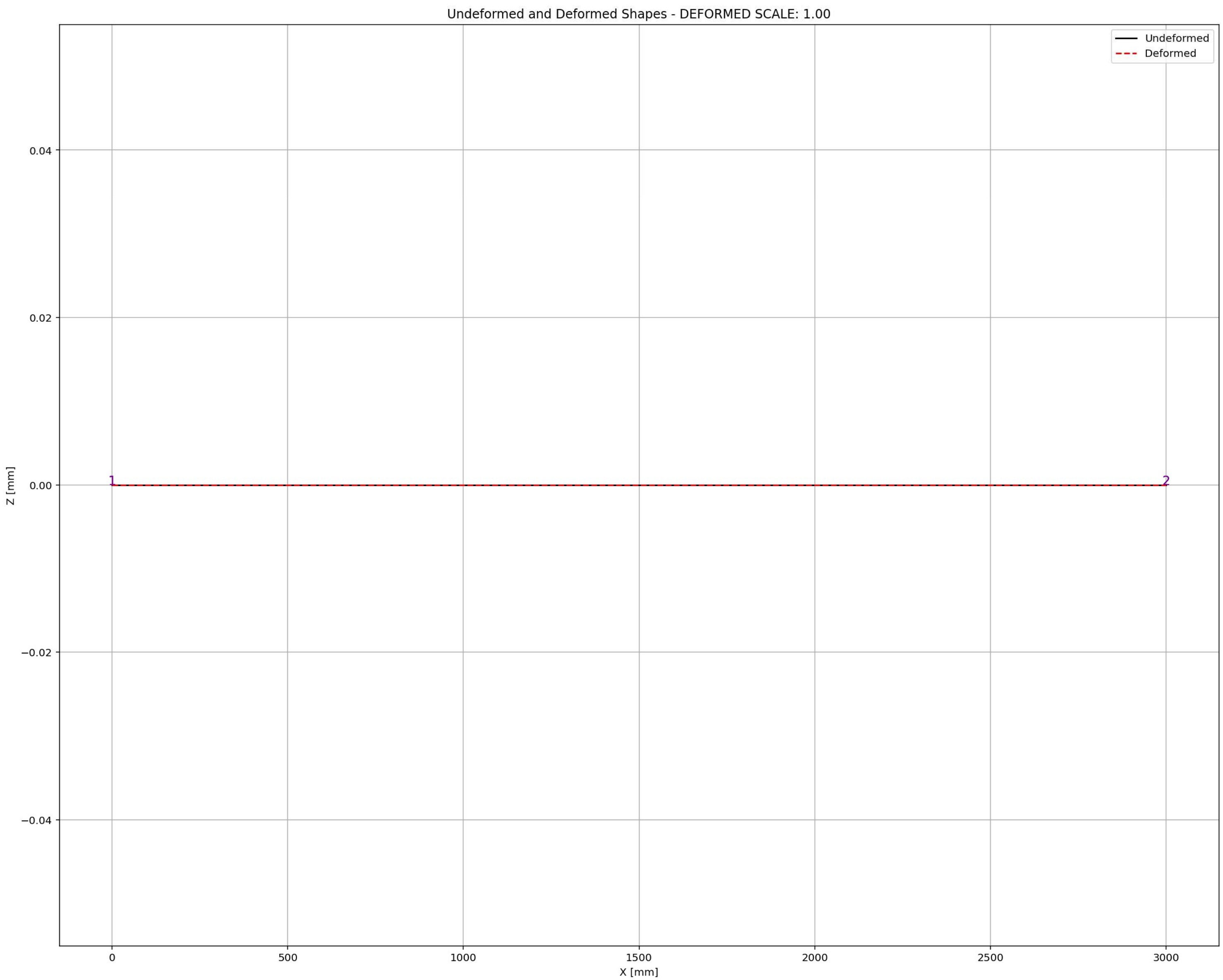


Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
1e6
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Base Reaction [N]
−40
−20
0
20
40
Displacement in Middle Span [mm]

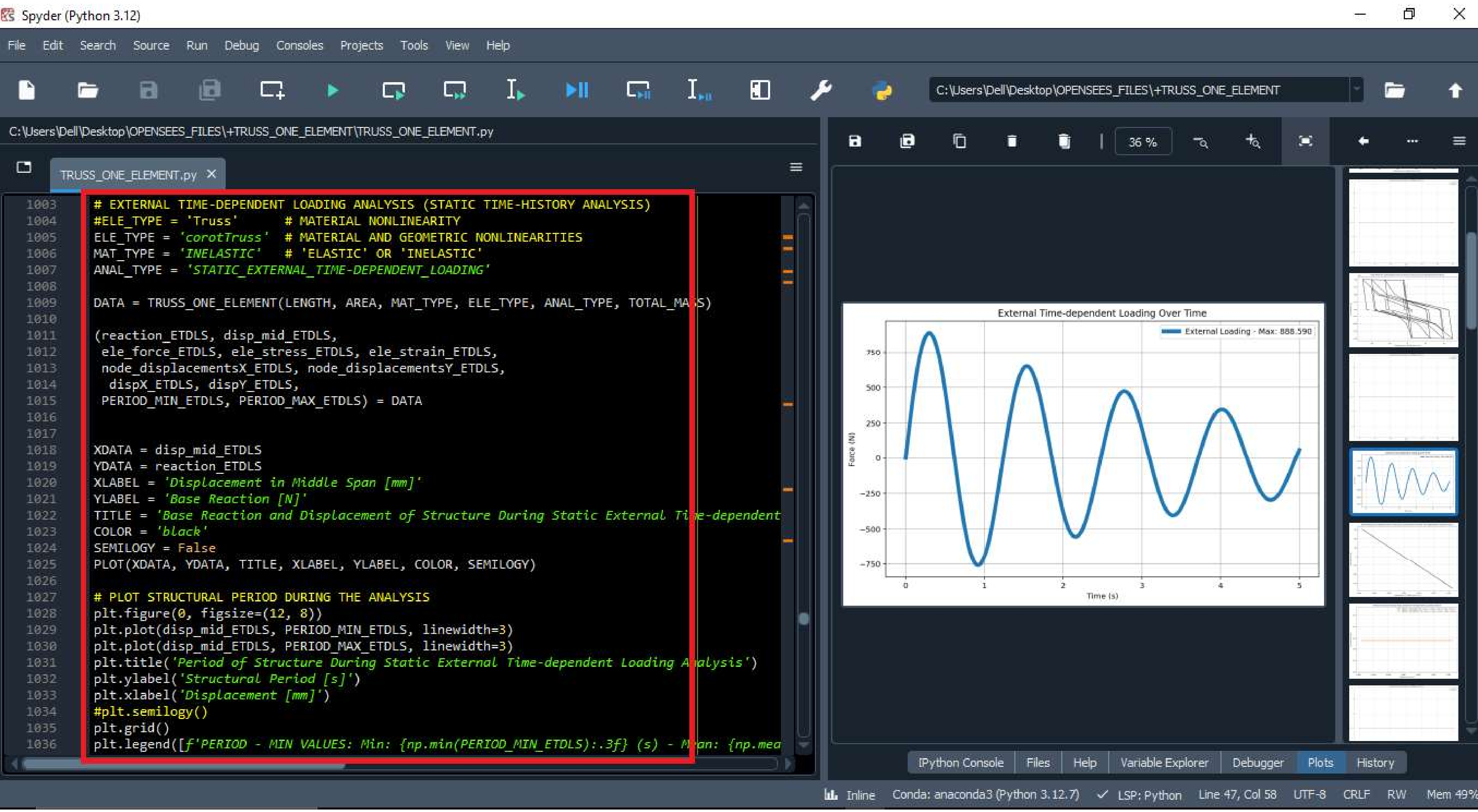

Period of Structure During Cyclic Displacement Analysis
Structural Period [s]
Displacement [mm]
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
−40
−20
0
20
40
PERIOD - MIN VALUES: Min: 0.084 (s) - Mean: 0.569 (s) - Max: 1.949 (s)
PERIOD - MAX VALUES: Min: 0.084 (s) - Mean: 0.569 (s) - Max: 1.949 (s)



Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
1e6
$\xi_{eq} = 42.8247$ [%]
$E_d = 134815960.586$ N·mm
$E_s = 25051703.608$ N·mm
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Base Reaction [N]
−40
−20
0
20
40
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 134815960.586 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.04
0.02
0.00
−0.02
−0.04
0
500
1000
1500
2000
2500
3000
1
2

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



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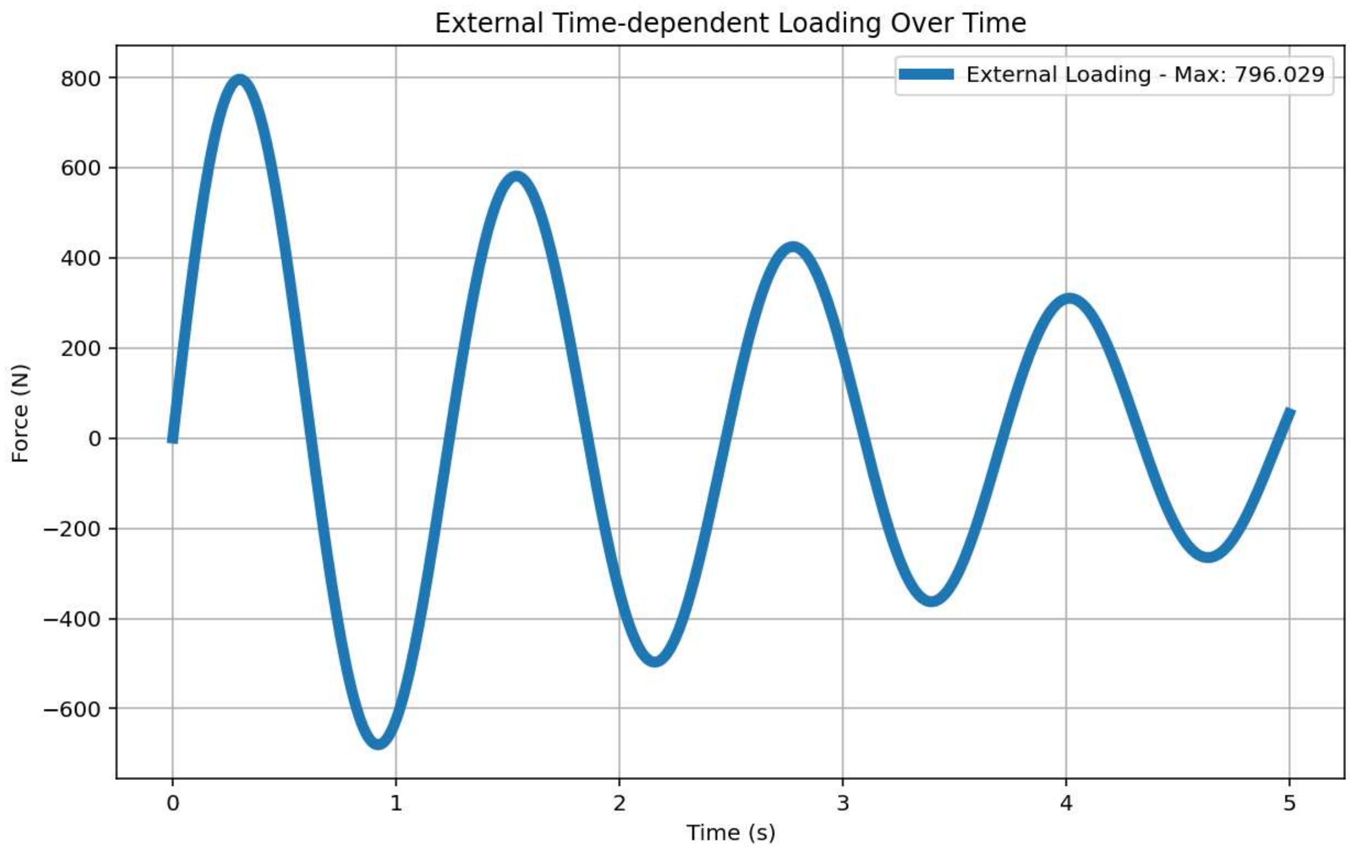
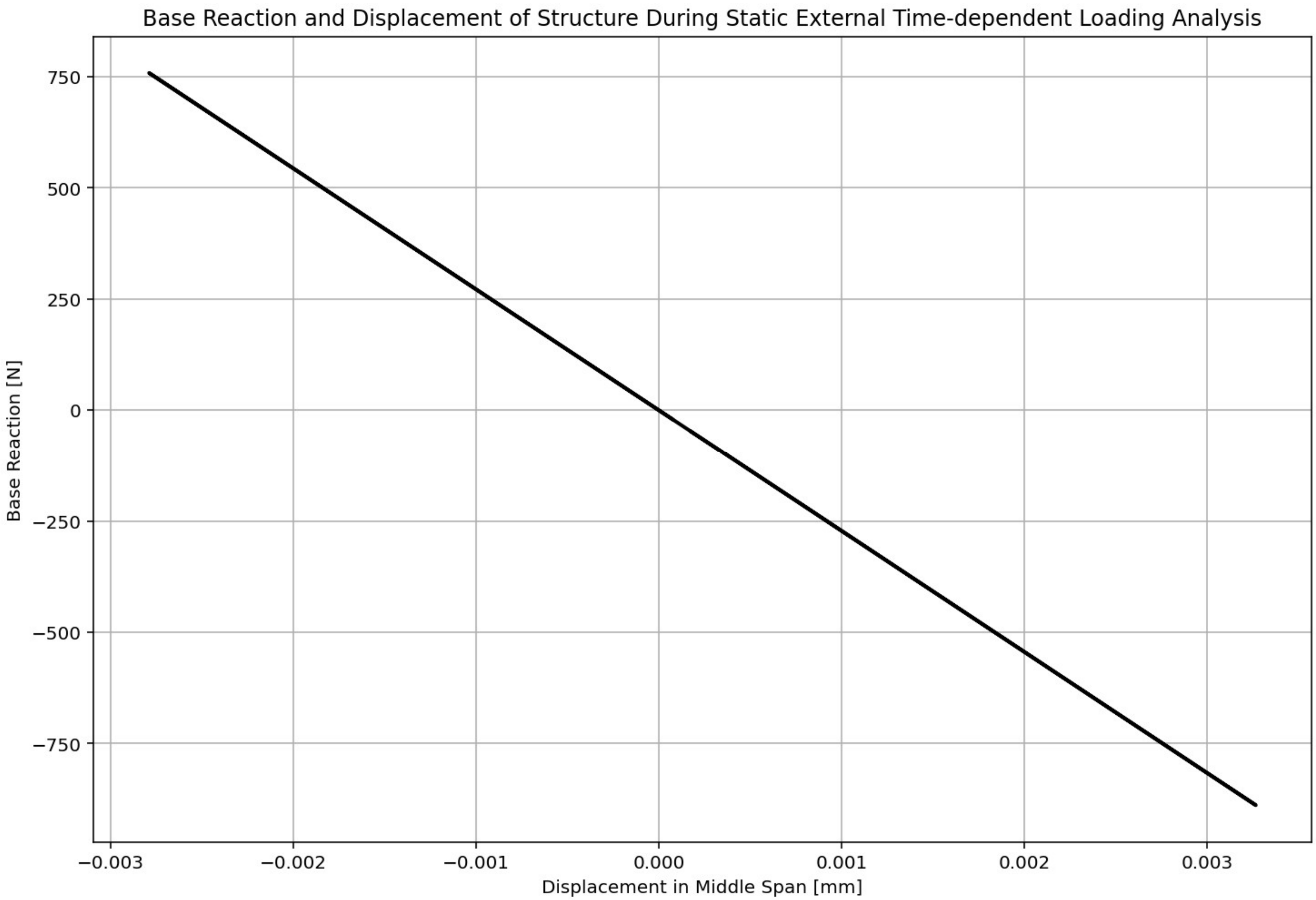
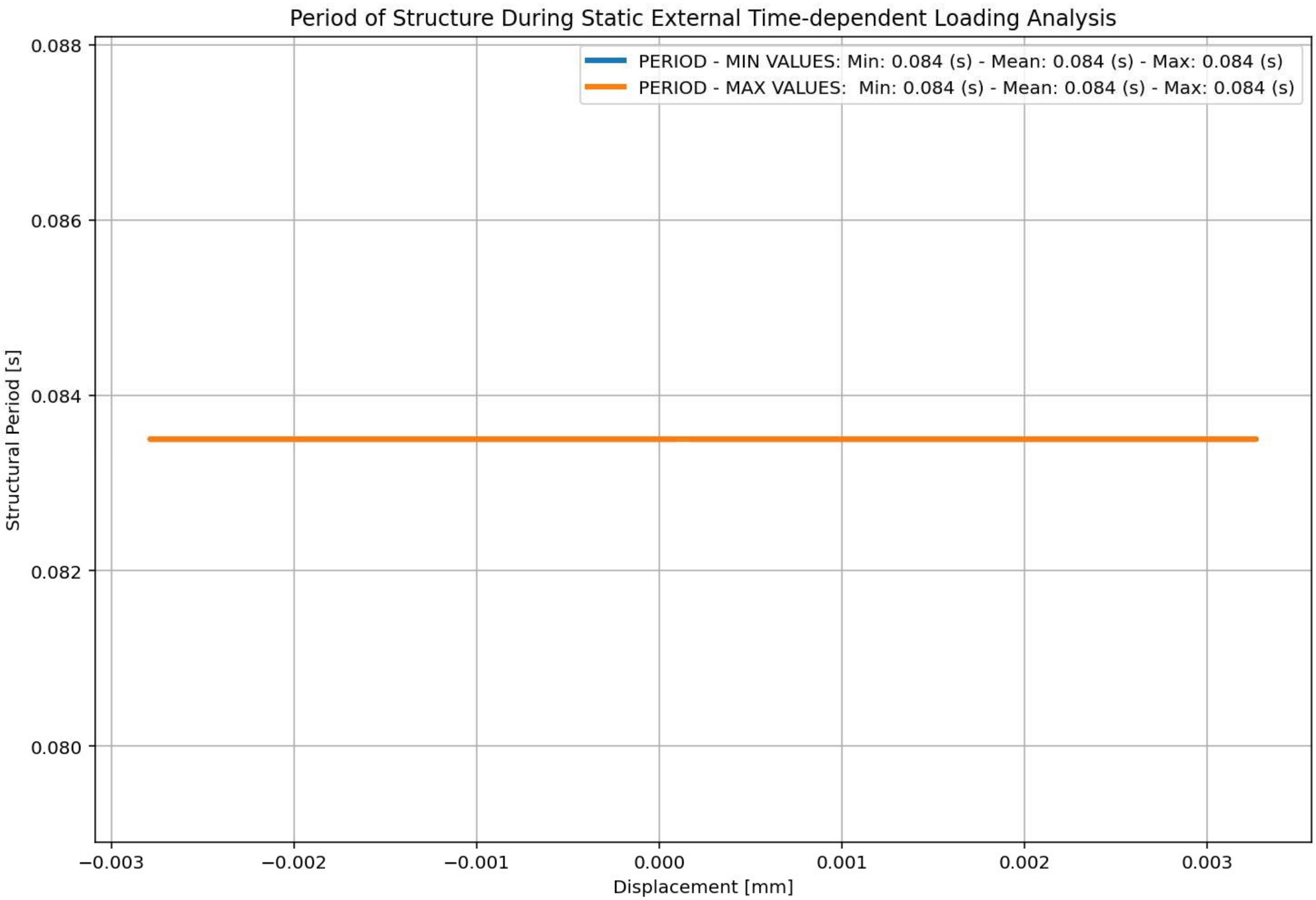
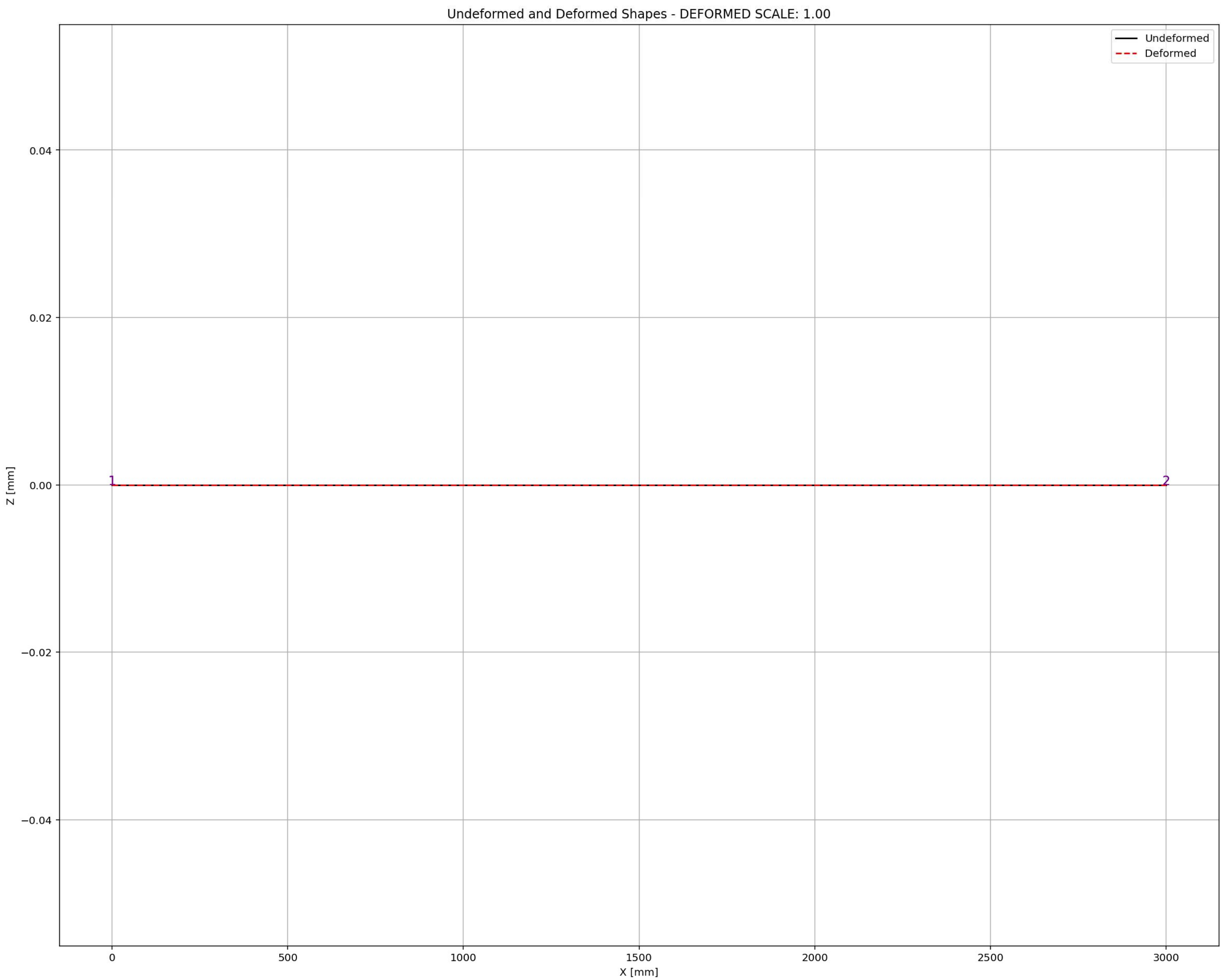
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reaction_ETDLS, disp_mid_ETDLS,
 ele_force_ETDLS, ele_stress_ETDLS, ele_strain_ETDLS,
 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
  dispX_ETDLS, dispY_ETDLS,
 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
XDATA = disp_mid_ETDLS
YDATA = reaction_ETDLS
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mea
36 %
External Time-dependent Loading Over Time
External Loading - Max: 888.590
Force (N)
Time (s)
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 49%


External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
Time (s)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5

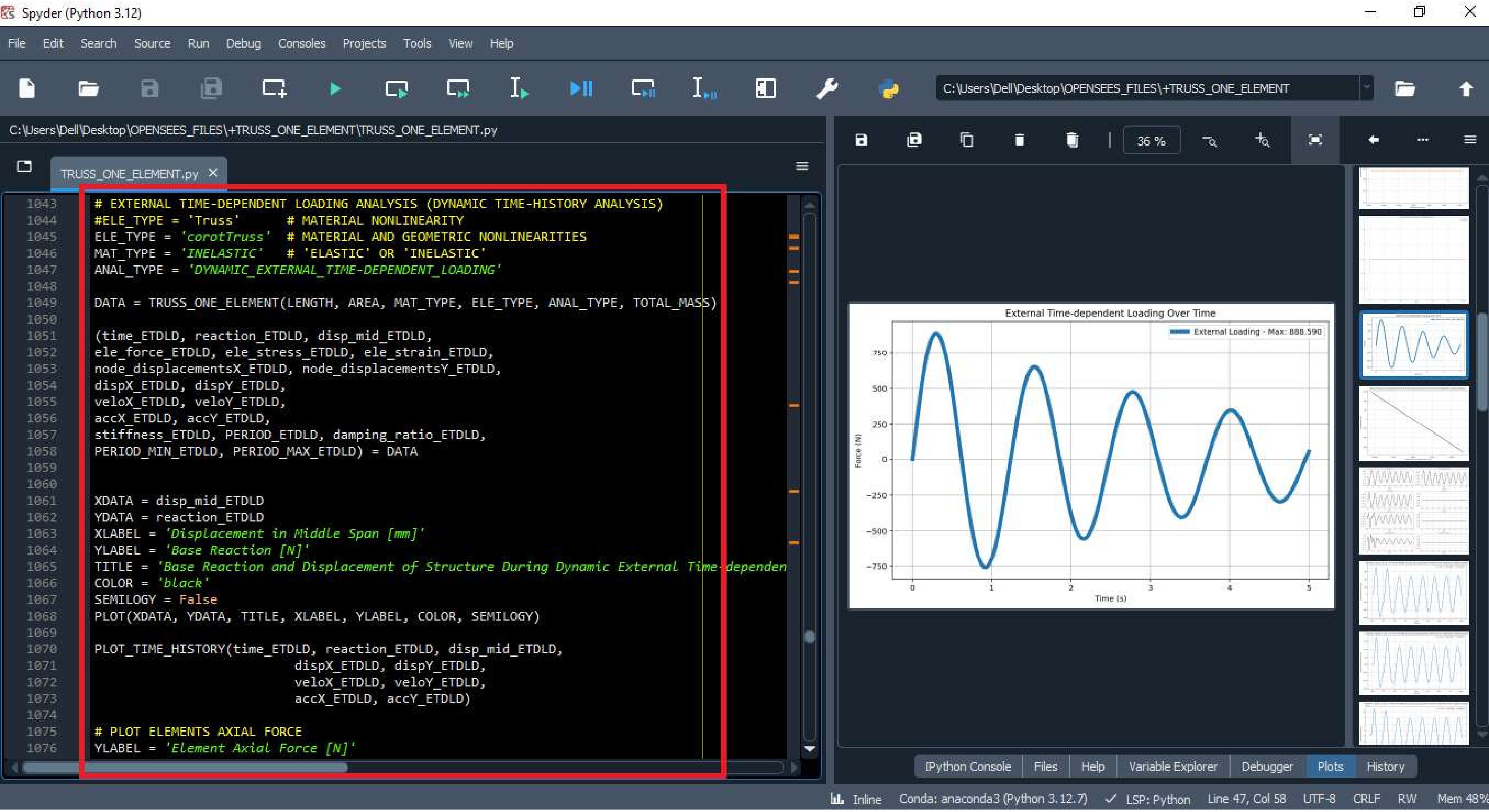

Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
750
500
250
0
−250
−500
−750
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement in Middle Span [mm]



Period of Structure During Static External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: 0.084 (s) - Mean: 0.084 (s) - Max: 0.084 (s)
PERIOD - MAX VALUES: Min: 0.084 (s) - Mean: 0.084 (s) - Max: 0.084 (s)
Structural Period [s]
0.088
0.086
0.084
0.082
0.080
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [mm]


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.04
0.02
0.00
−0.02
−0.04
0
500
1000
1500
2000
2500
3000
1
2

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



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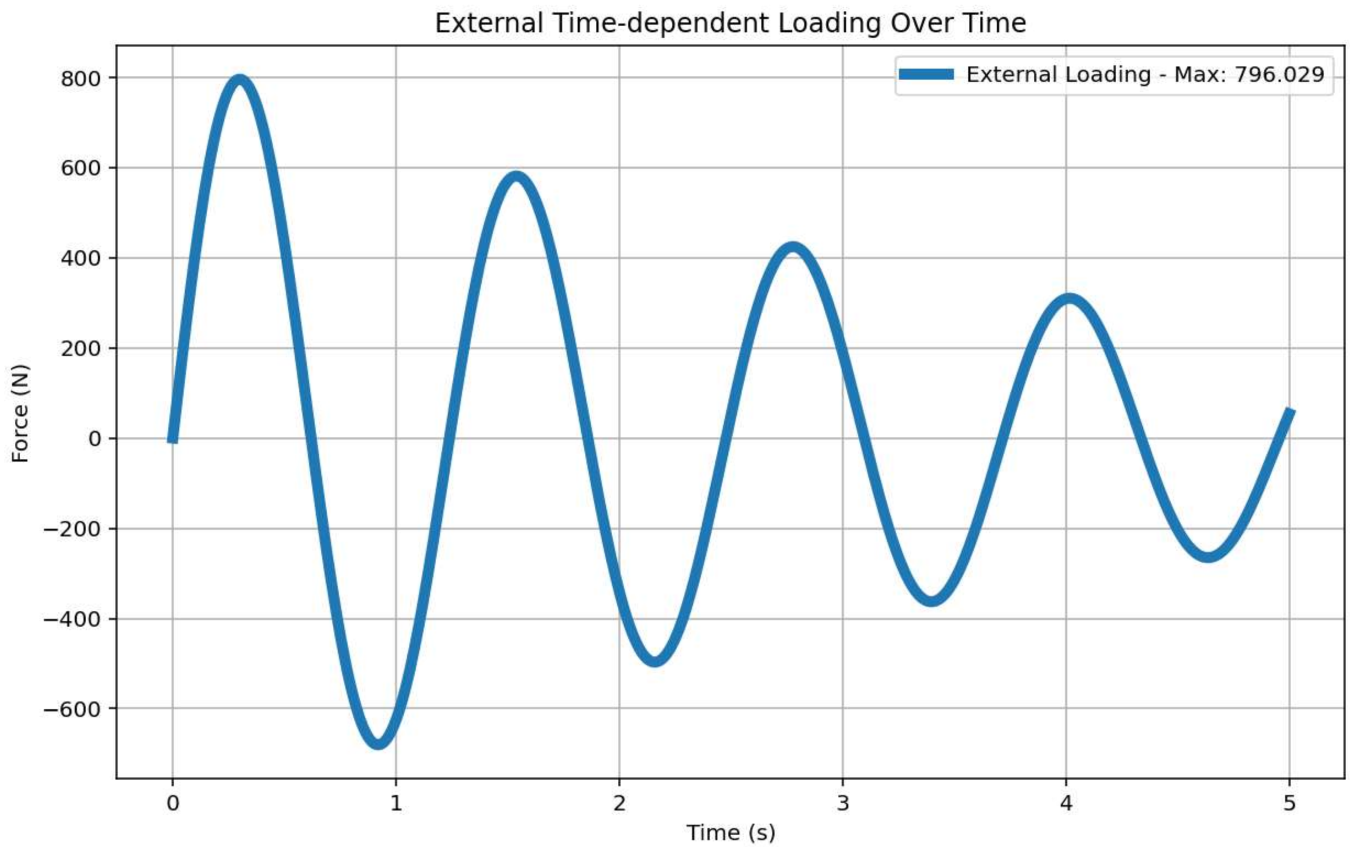
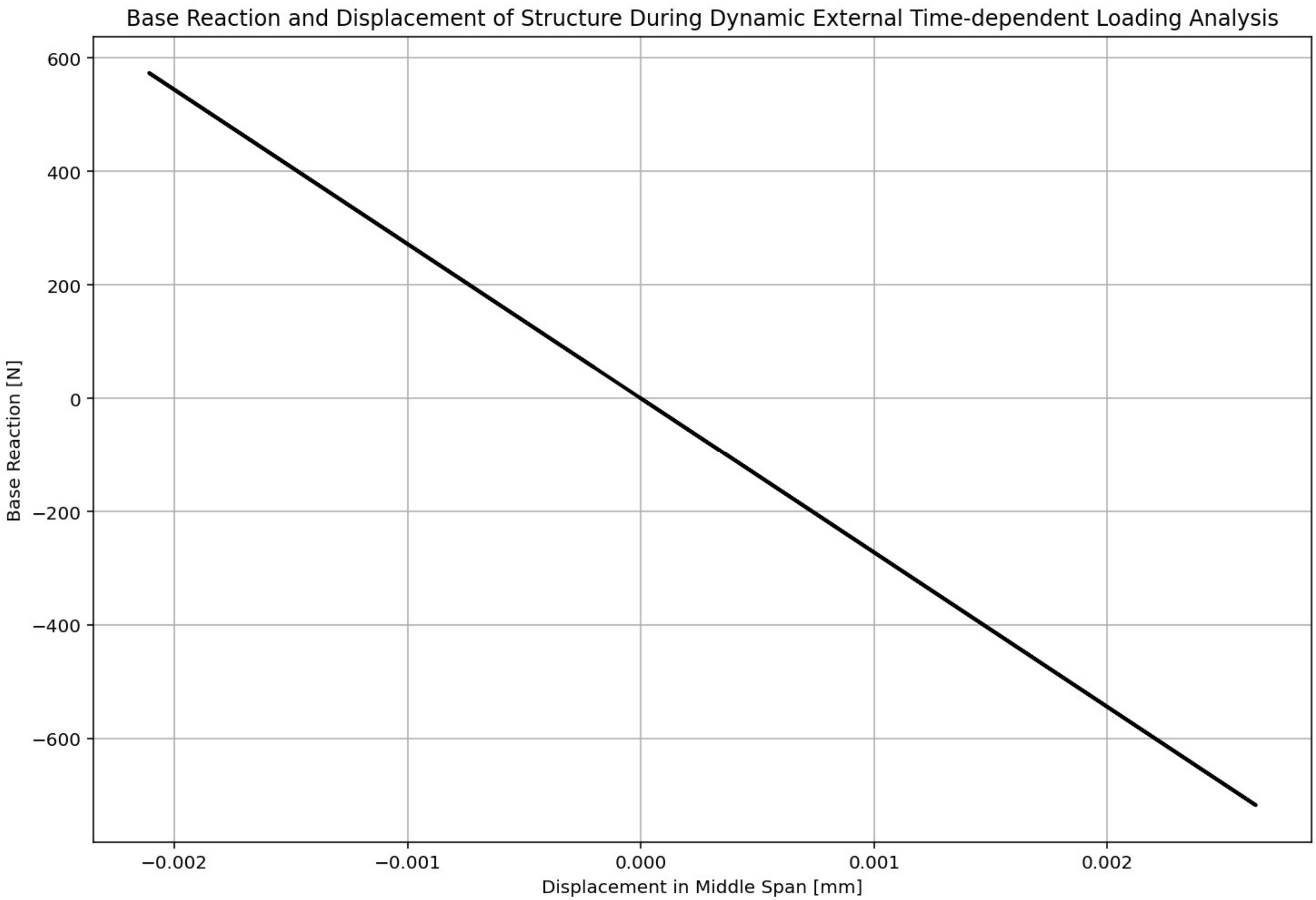
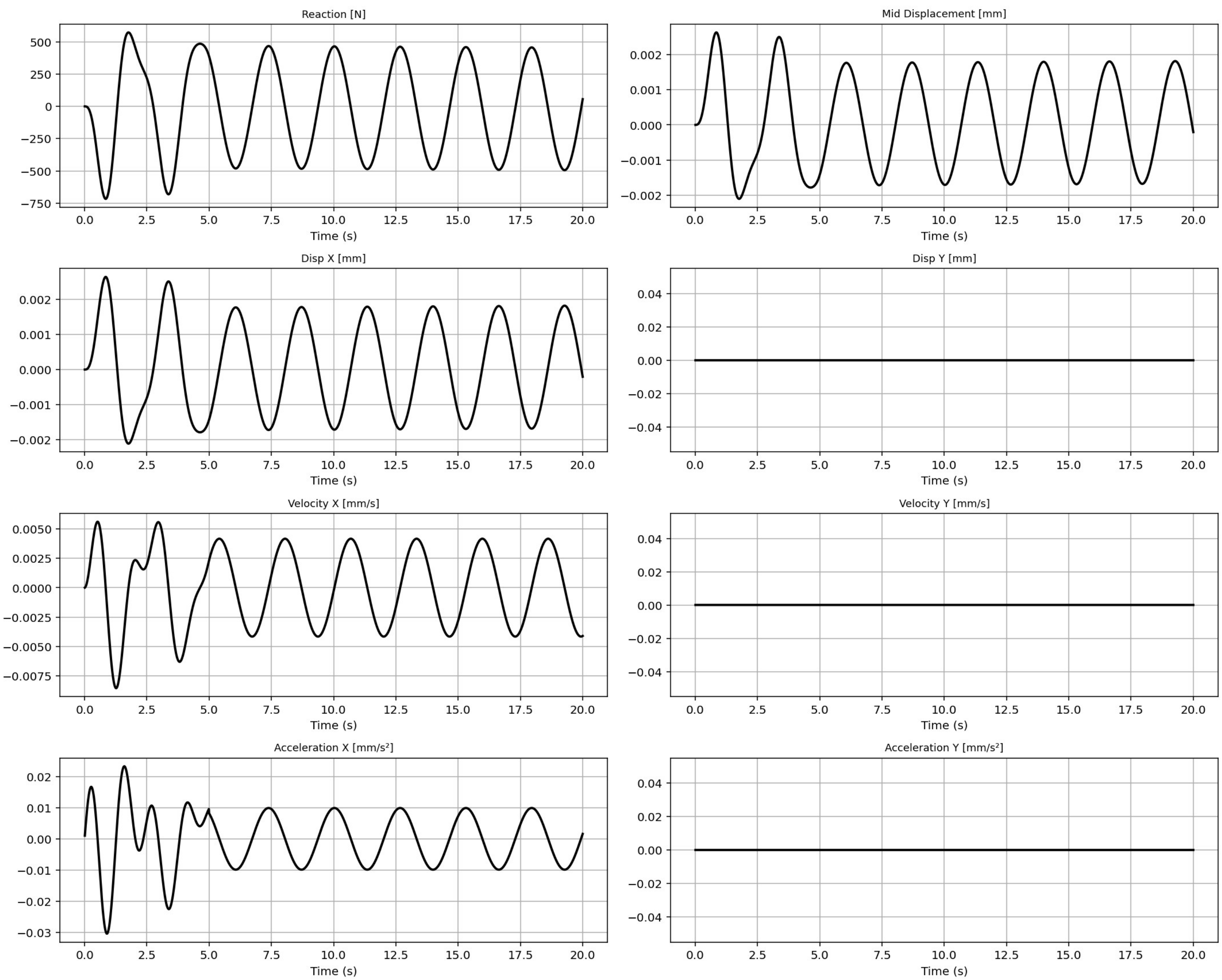
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
ele_force_ETDLD, ele_stress_ETDLD, ele_strain_ETDLD,
node_displacementsX_ETDLD, node_displacementsY_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD,
stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
XDATA = disp_mid_ETDLD
YDATA = reaction_ETDLD
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependen
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
                  dispX_ETDLD, dispY_ETDLD,
                  veloX_ETDLD, veloY_ETDLD,
                  accX_ETDLD, accY_ETDLD)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
36 %
External Time-dependent Loading Over Time
External Loading - Max: 888.590
Force (N)
Time (s)
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 48%

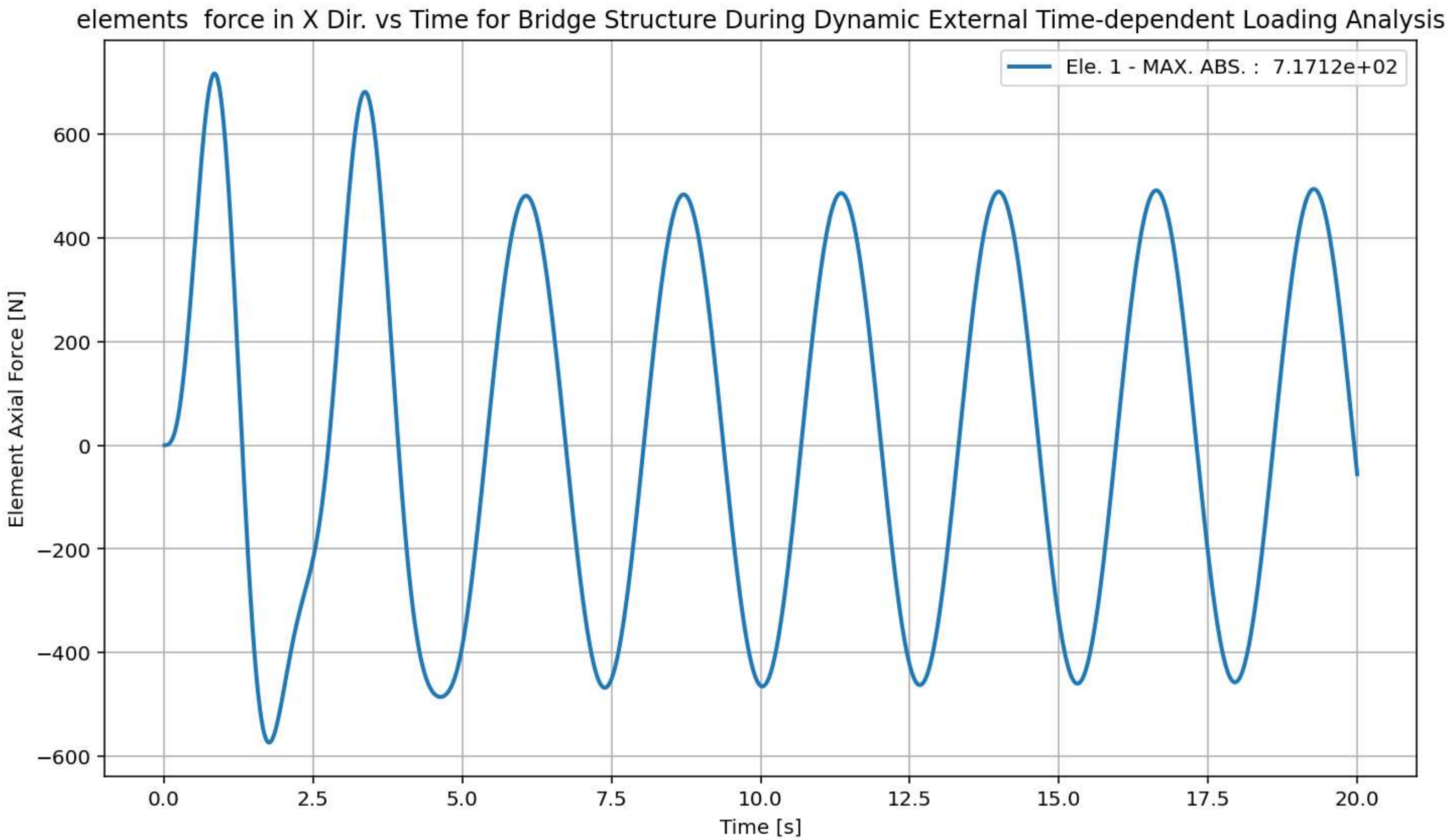


External Time-dependent Loading Over Time
External Loading - Max: 796.029
800
600
400
200
0
−200
−400
−600
Force (N)
0
1
2
3
4
5
Time (s)

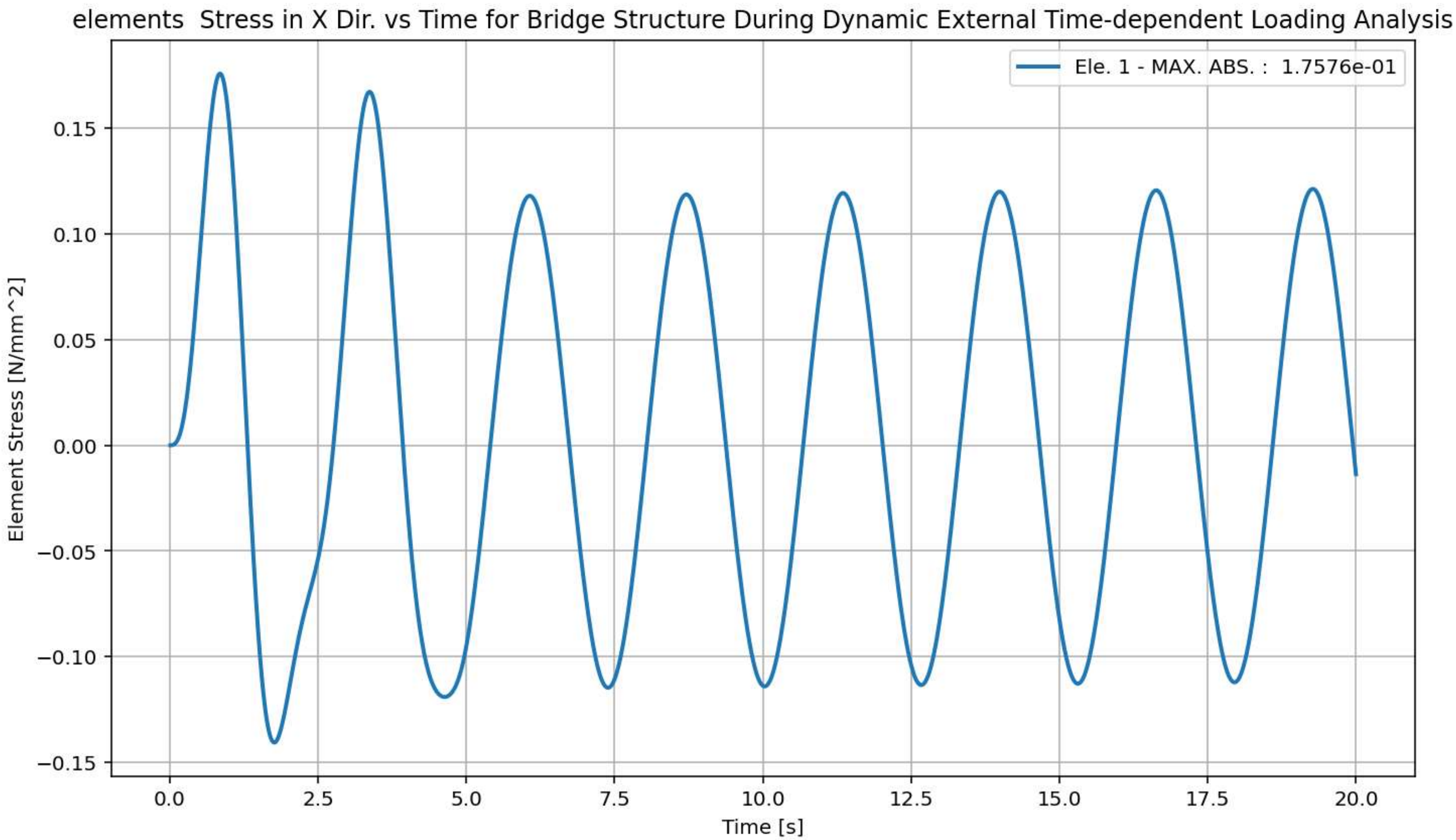

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
600
400
200
0
−200
−400
−600
−0.002
−0.001
0.000
0.001
0.002
Displacement in Middle Span [mm]



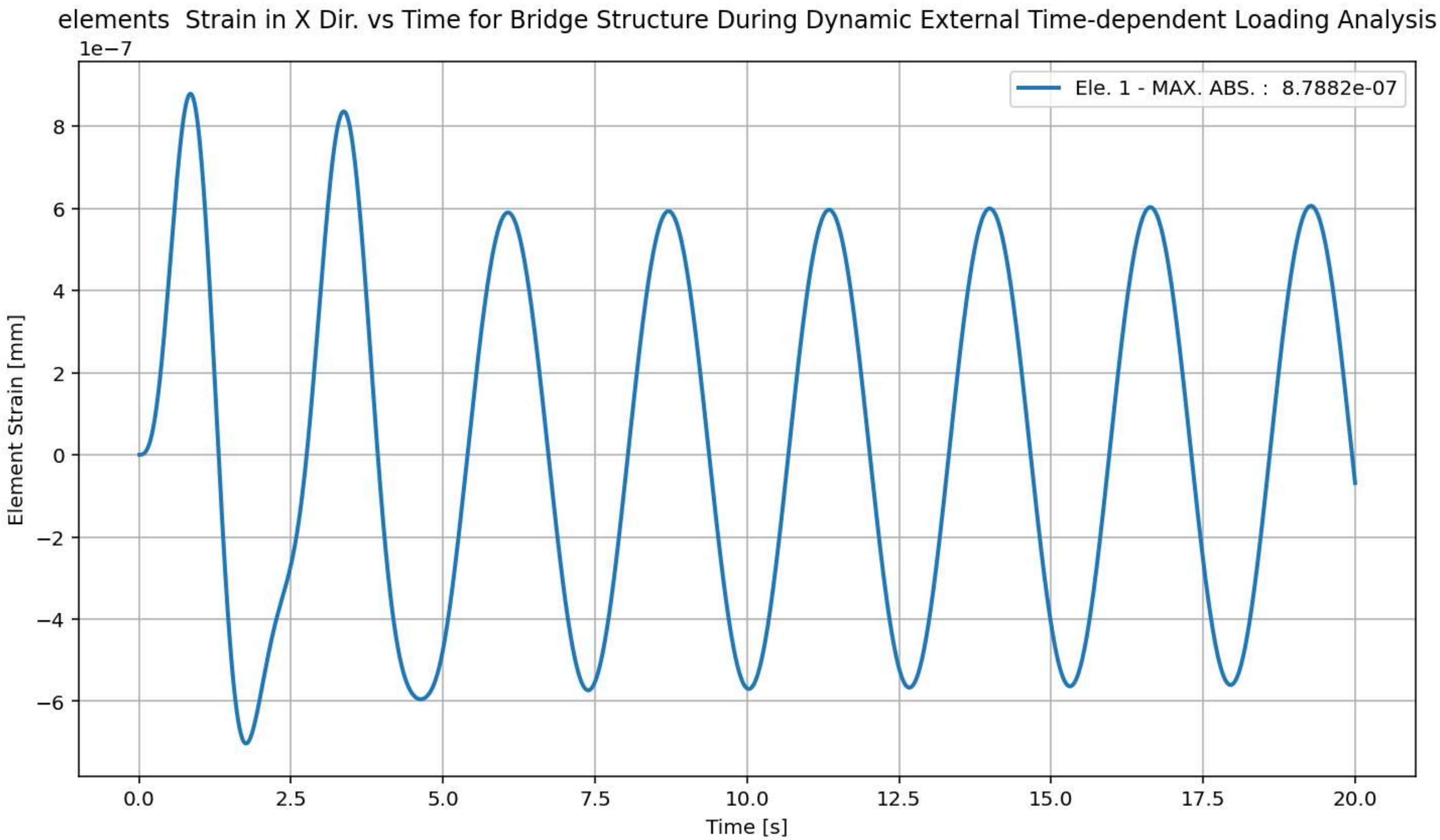
Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

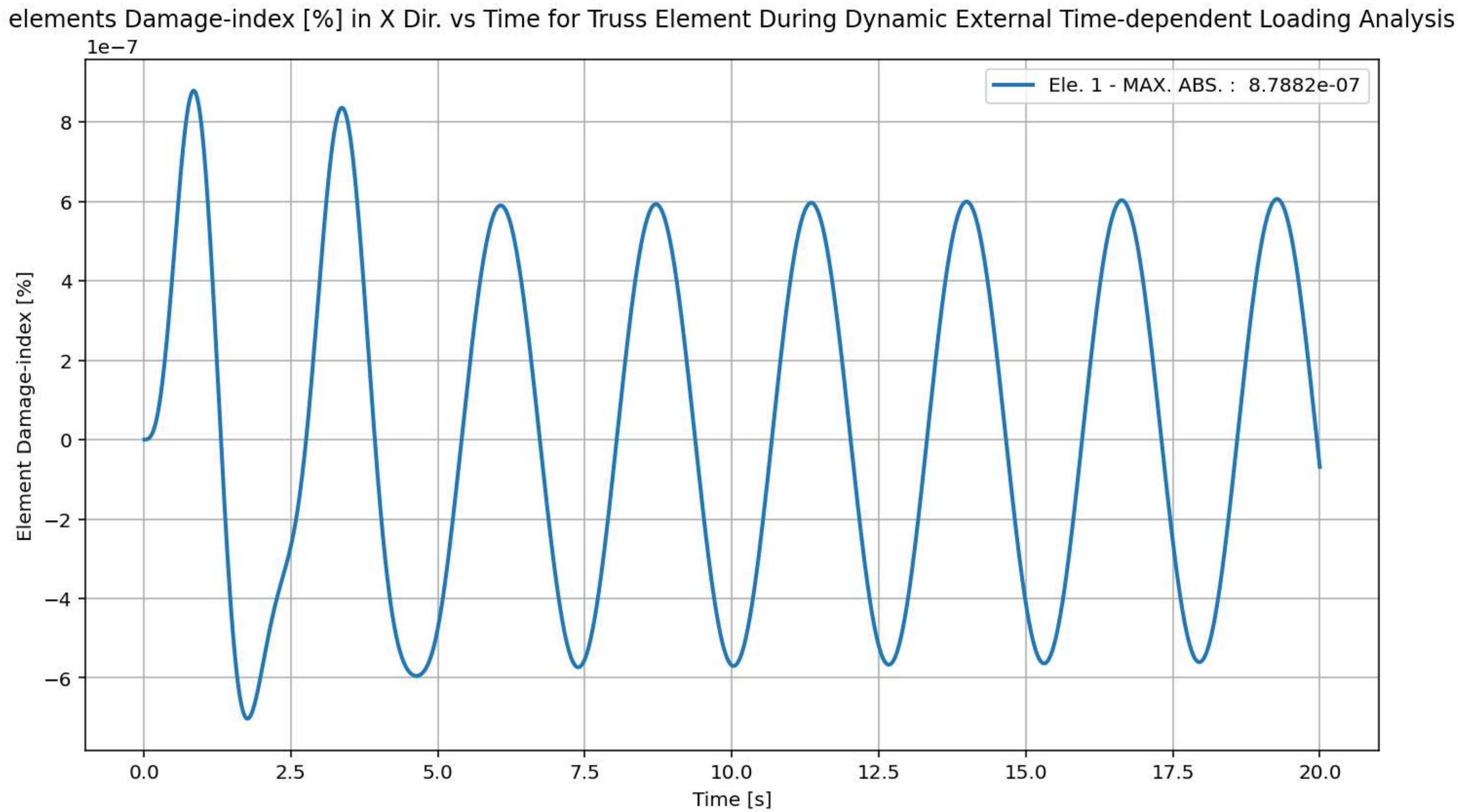


elements force in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 7.1712e+02
Element Axial Force [N]
600
400
200
0
−200
−400
−600
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

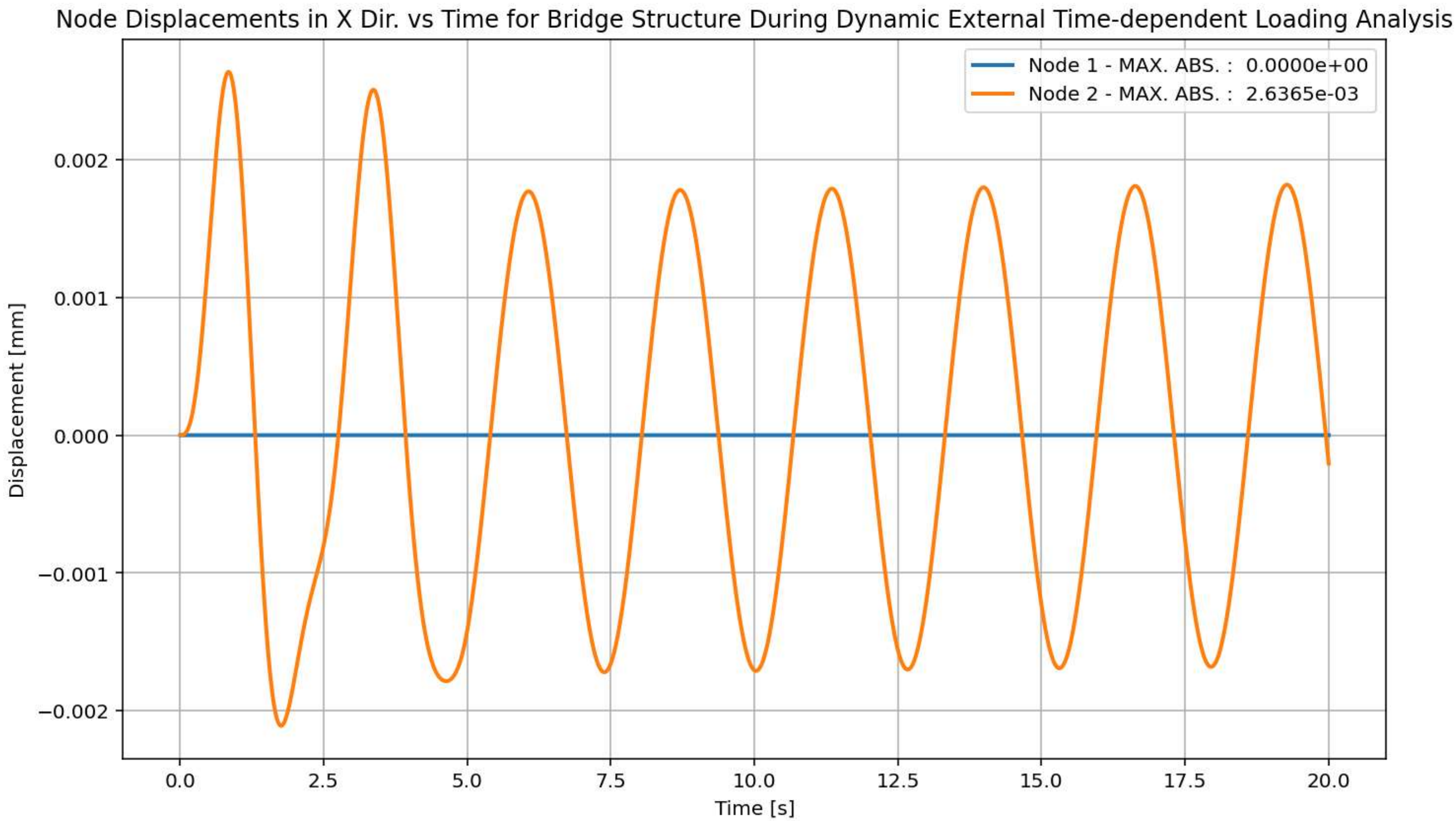


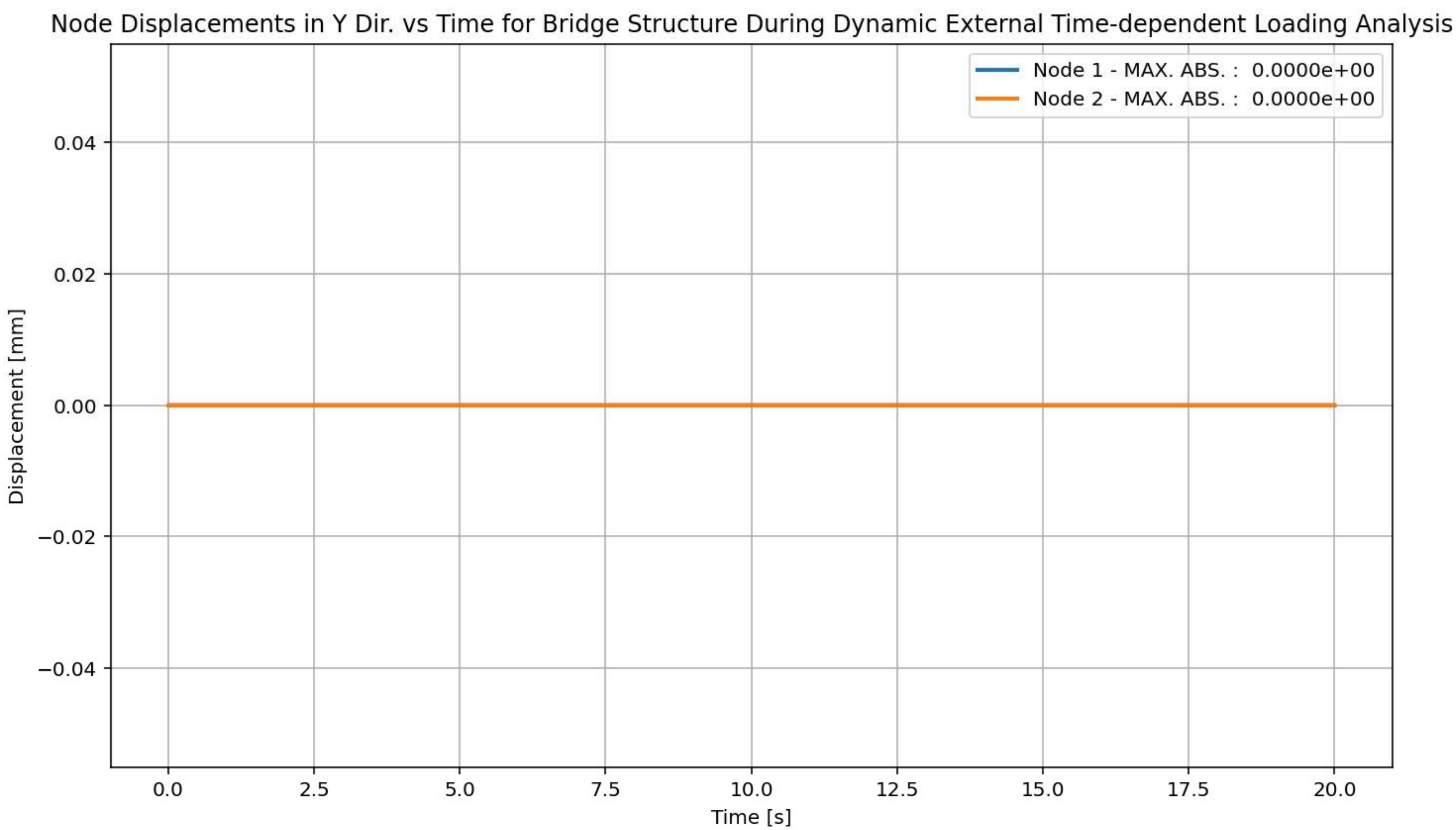
elements Stress in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 1.7576e-01
Element Stress [N/mm^2]
Time [s]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


elements Strain in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
1e−7
Ele. 1 - MAX. ABS. : 8.7882e-07
Element Strain [mm]
Time [s]
8
6
4
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

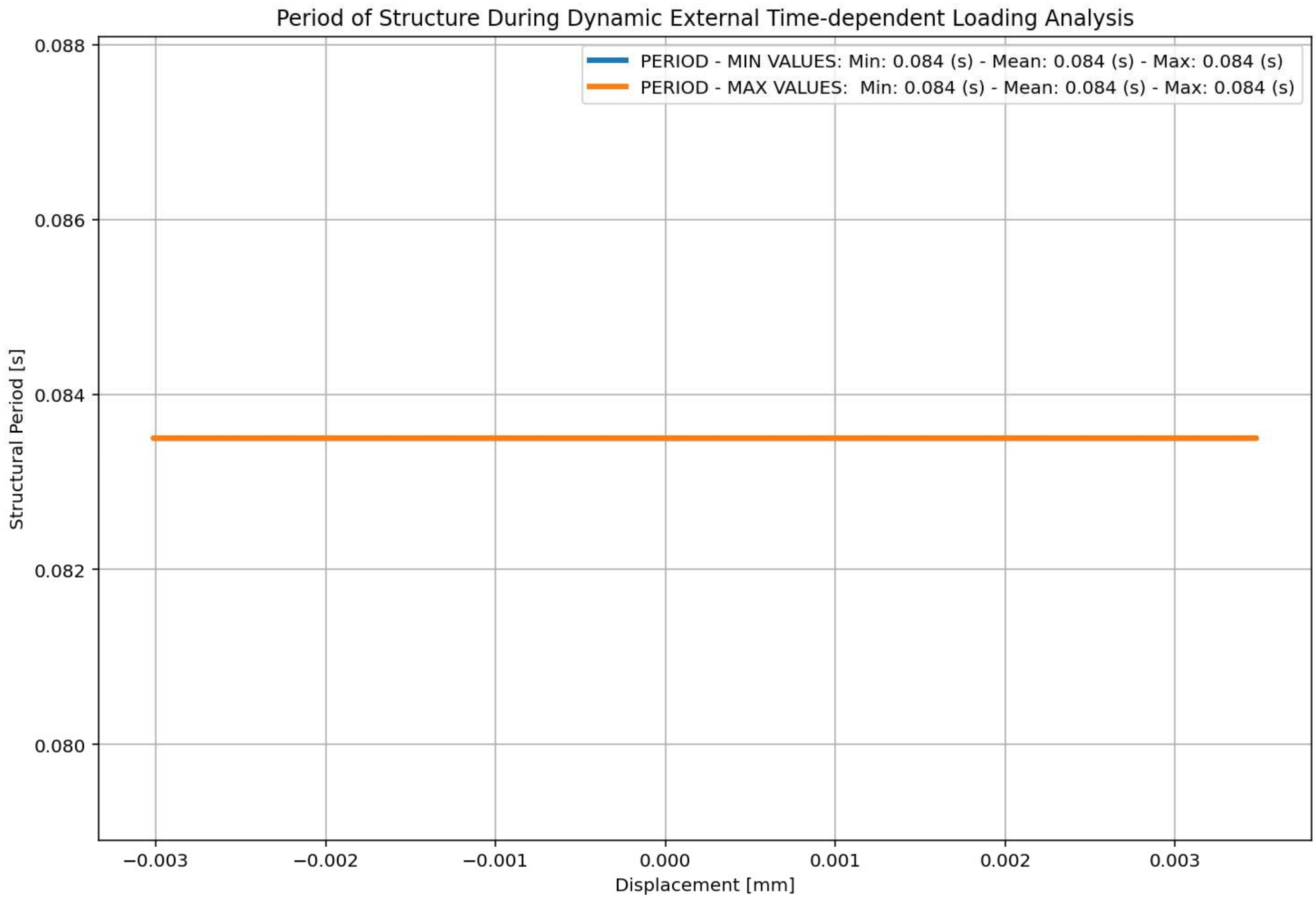


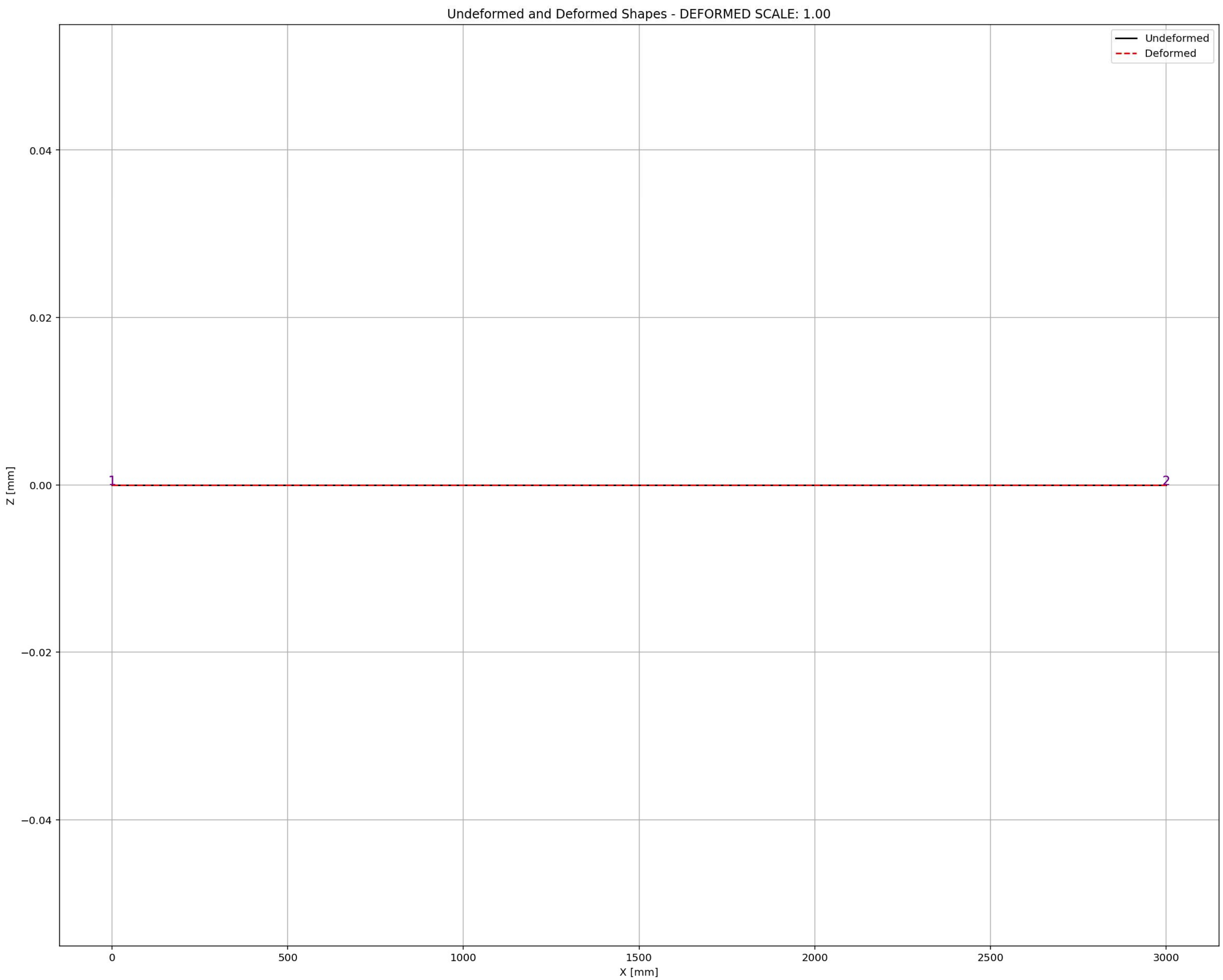
elements Damage-index [%] in X Dir. vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis
1e−7
Ele. 1 - MAX. ABS. : 8.7882e-07
Element Damage-index [%]
Time [s]
8
6
4
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Node Displacements in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.6365e-03
Displacement [mm]
Time [s]
0.002
0.001
0.000
−0.001
−0.002
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

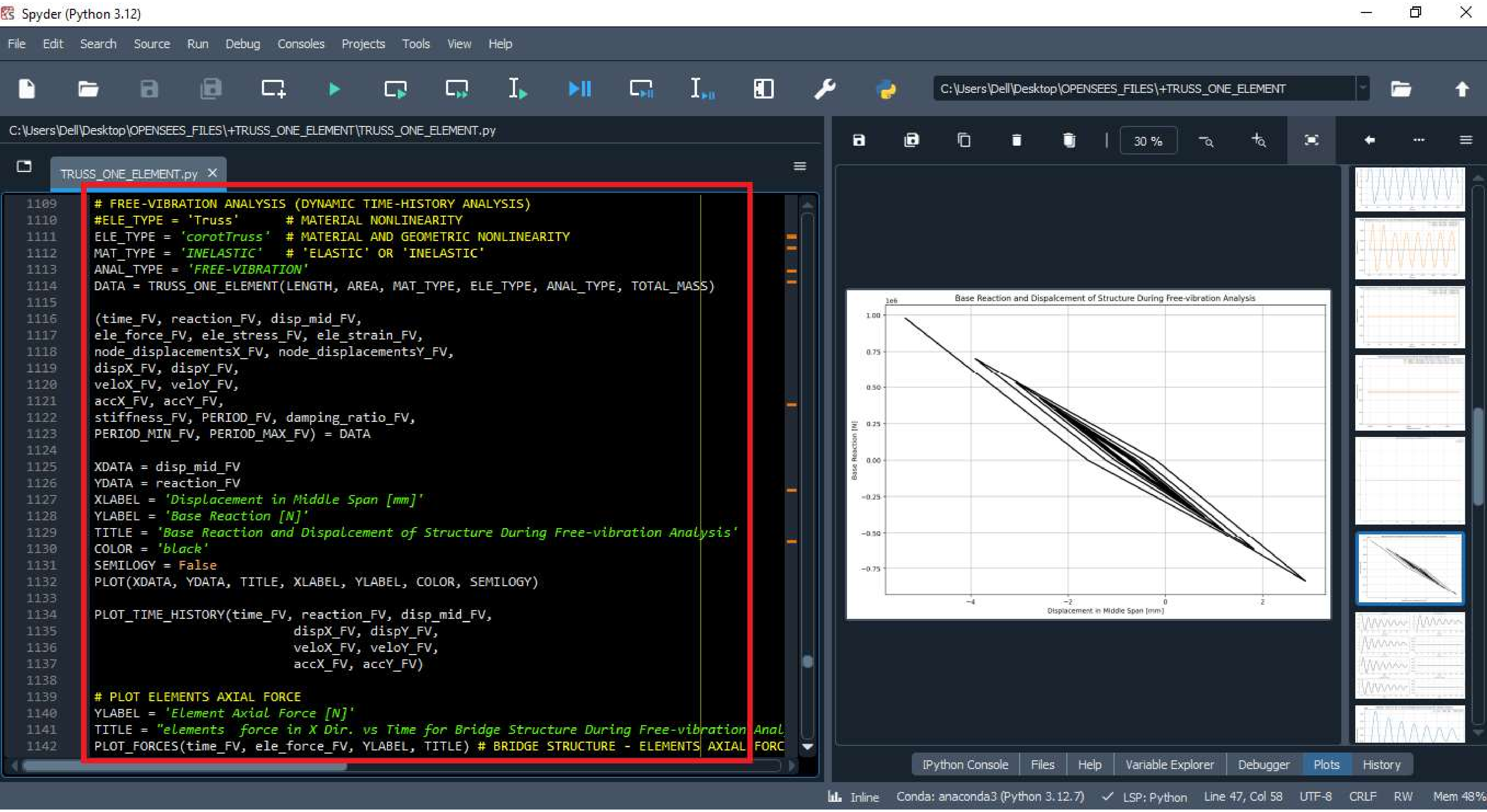


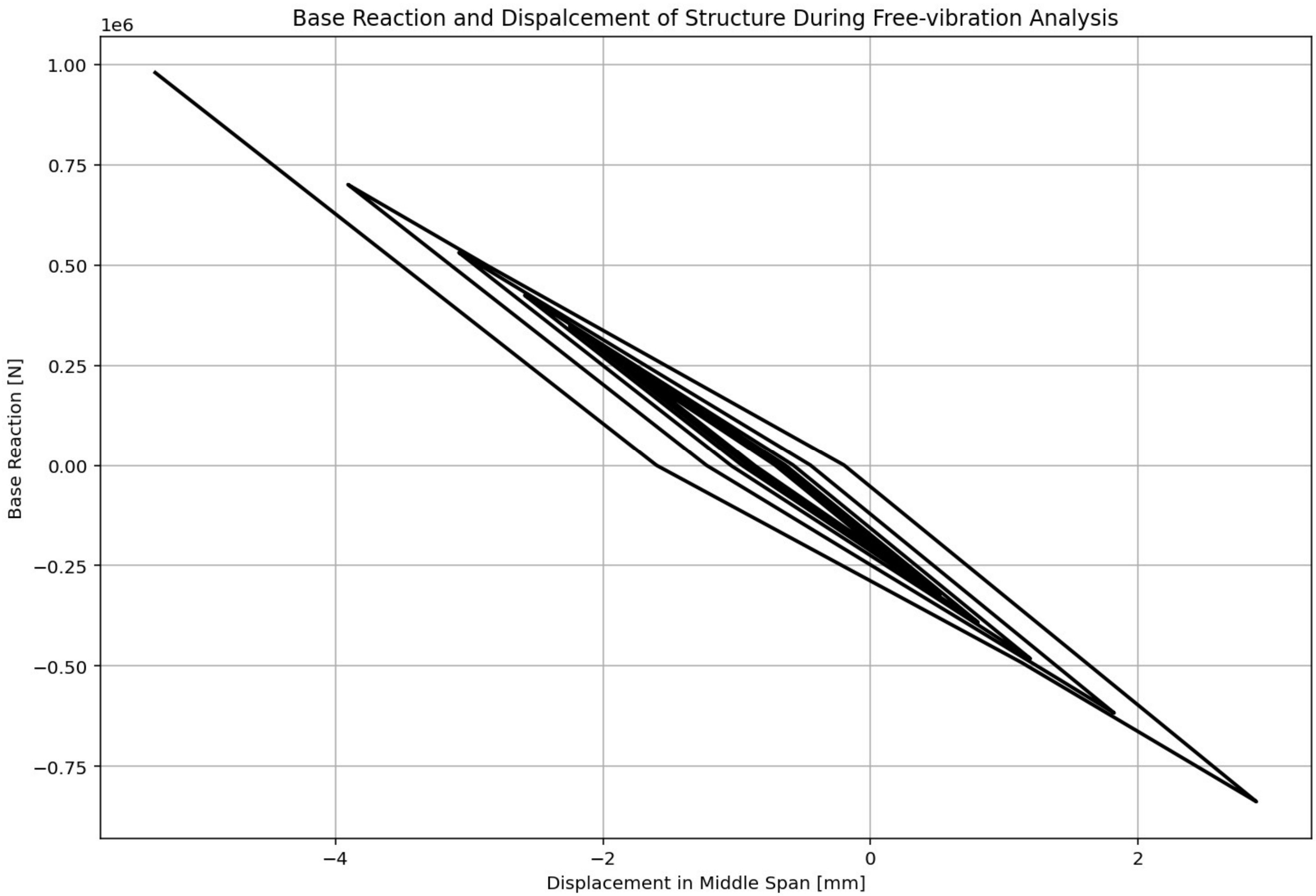
Node Displacements in Y Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 0.0000e+00
0.04
0.02
0.00
−0.02
−0.04
Displacement [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Period of Structure During Dynamic External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: 0.084 (s) - Mean: 0.084 (s) - Max: 0.084 (s)
PERIOD - MAX VALUES: Min: 0.084 (s) - Mean: 0.084 (s) - Max: 0.084 (s)
Structural Period [s]
0.088
0.086
0.084
0.082
0.080
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [mm]

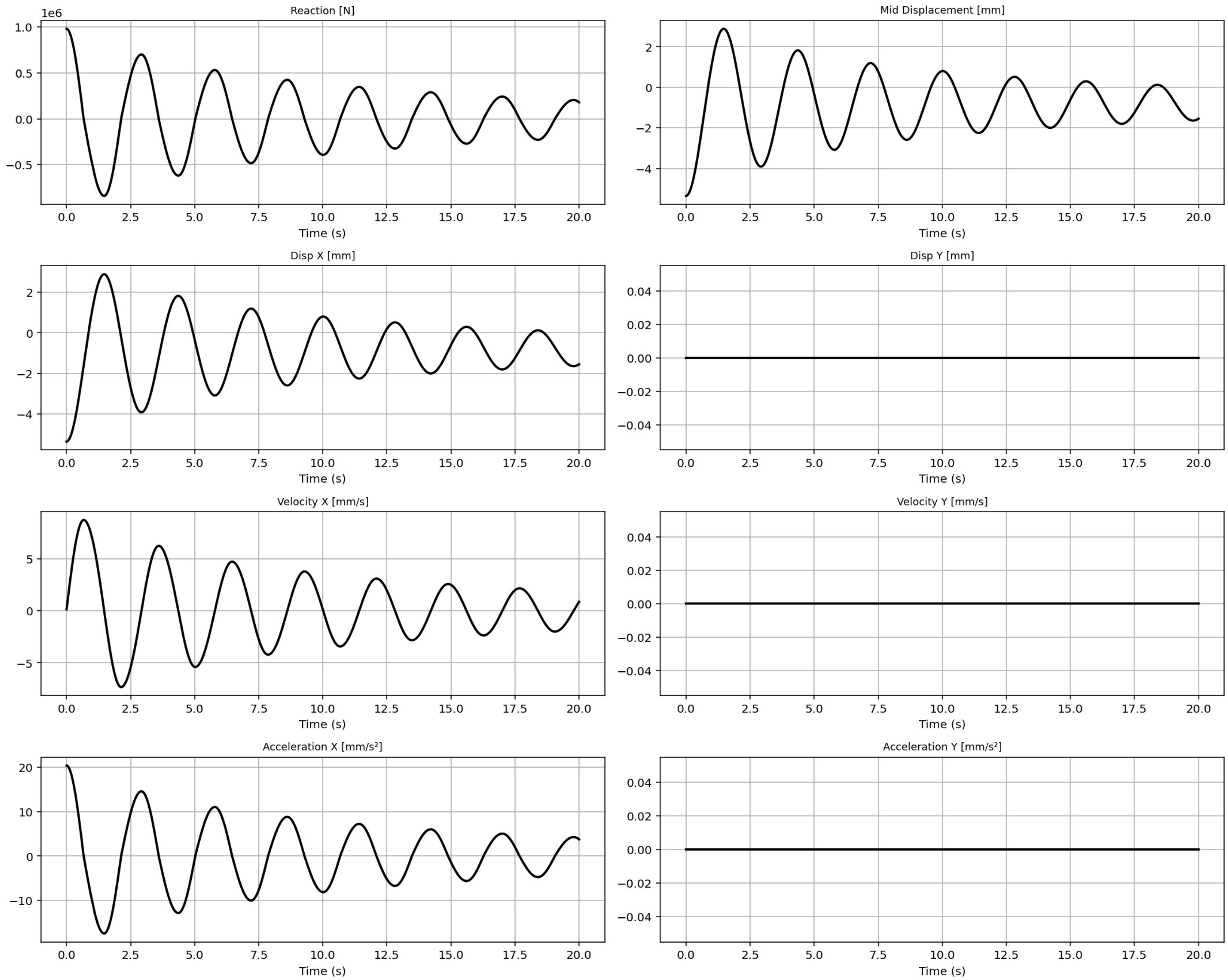

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.04
0.02
0.00
−0.02
−0.04
0
500
1000
1500
2000
2500
3000
1
2

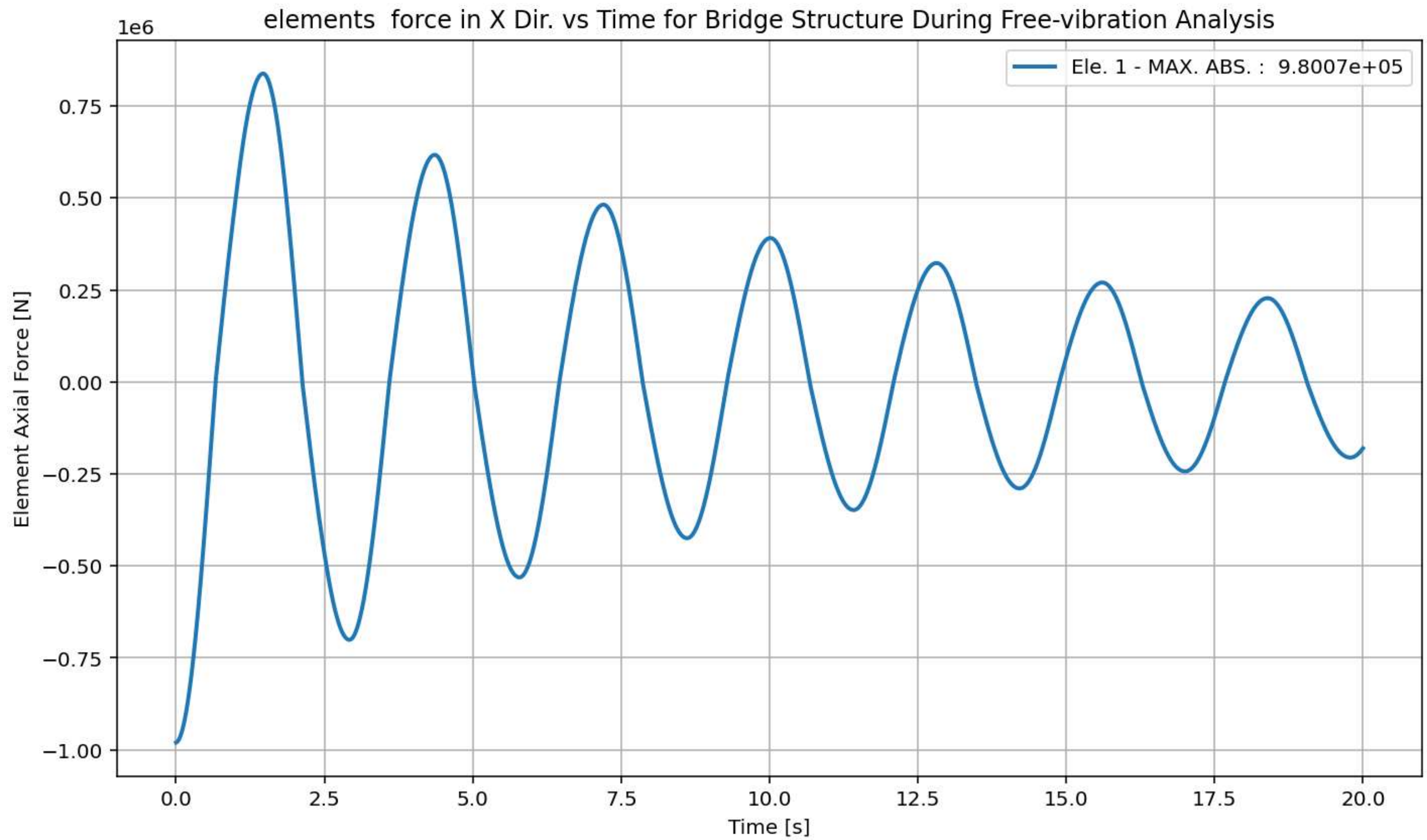
FREE-VIBRATION ANALYSIS

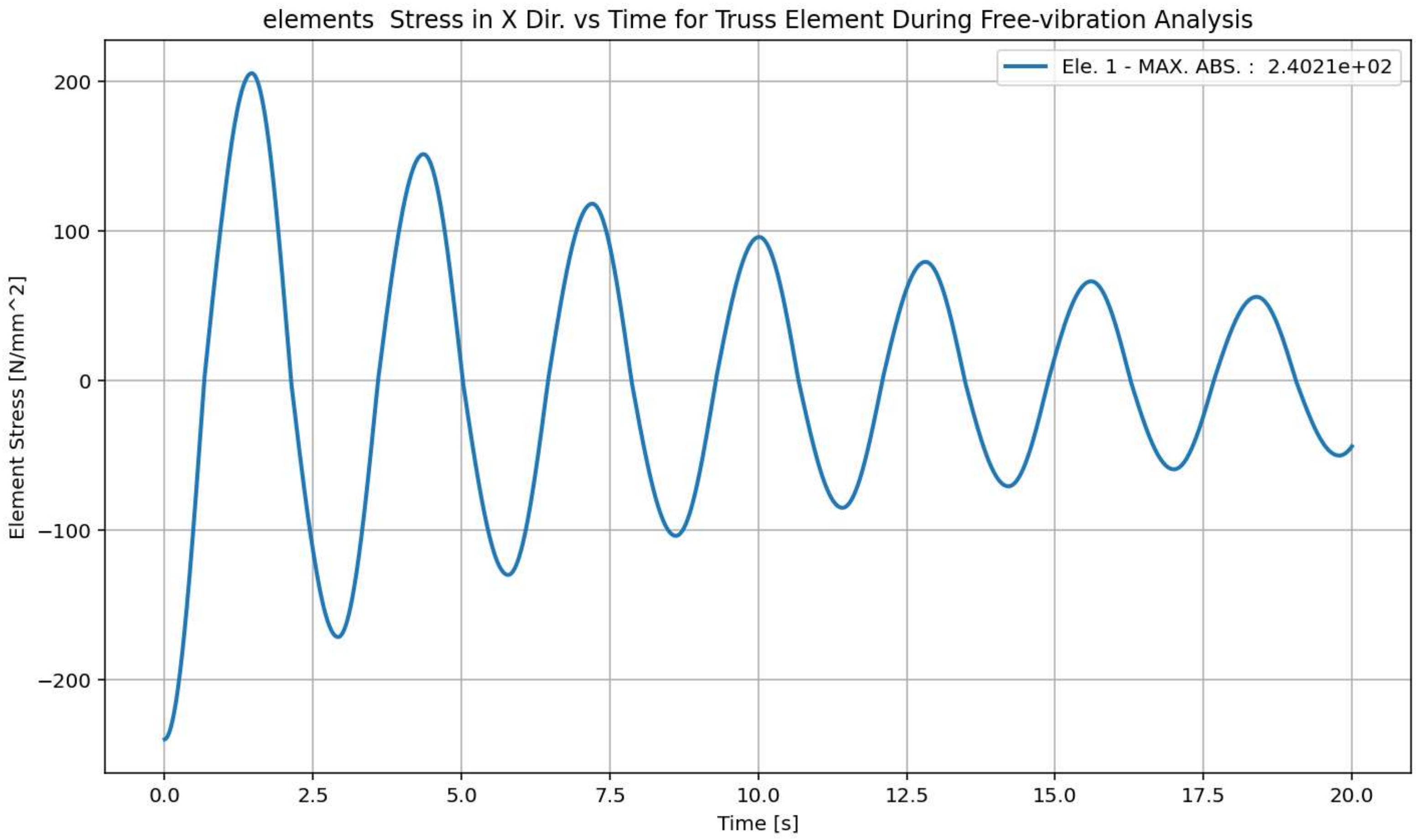

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITY
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'FREE-VIBRATION'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(time_FV, reaction_FV, disp_mid_FV,
ele_force_FV, ele_stress_FV, ele_strain_FV,
node_displacementsX_FV, node_displacementsY_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV,
stiffness_FV, PERIOD_FV, damping_ratio_FV,
PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
XDATA = disp_mid_FV
YDATA = reaction_FV
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
TITLE = "elements force in X Dir. vs Time for Bridge Structure During Free-vibration Anal
PLOT_FORCES(time_FV, ele_force_FV, YLABEL, TITLE) # BRIDGE STRUCTURE - ELEMENTS AXIAL FORC
30 %
Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 48%



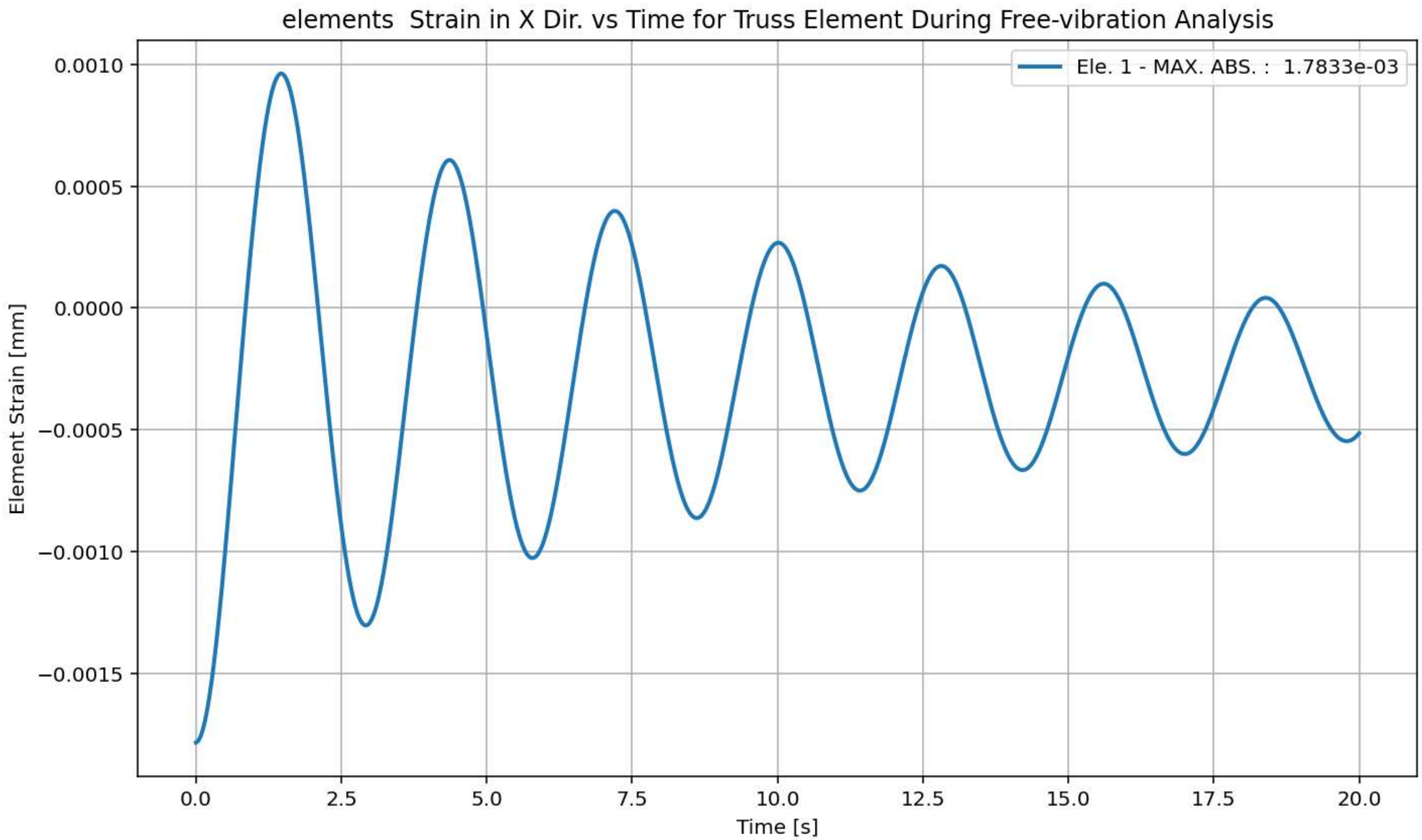
Base Reaction and Dispalcement of Structure During Free-vibration Analysis
1e6
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Base Reaction [N]
−4
−2
0
2
Displacement in Middle Span [mm]


Reaction [N]
1e6
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

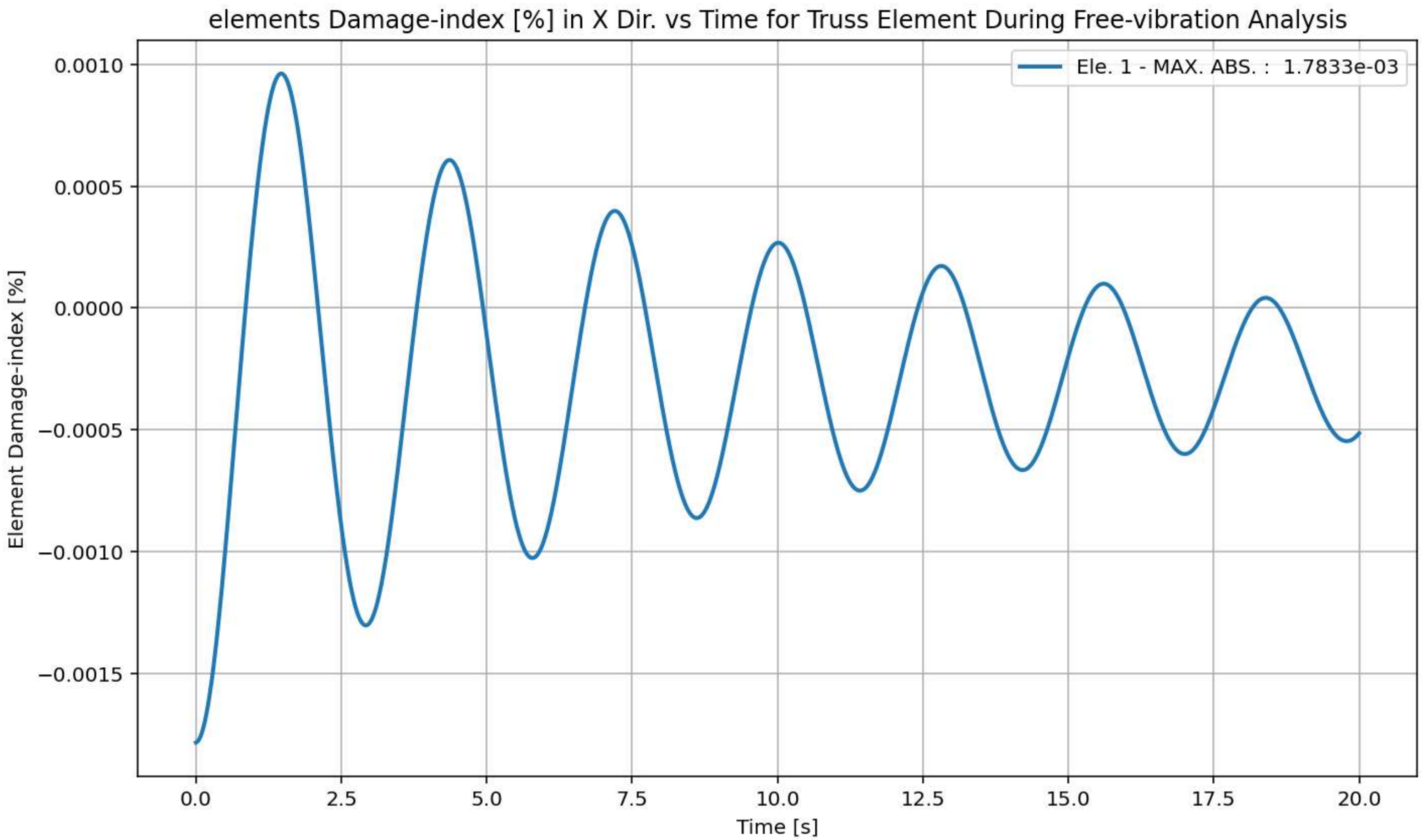

elements force in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
1e6
Ele. 1 - MAX. ABS. : 9.8007e+05
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Element Axial Force [N]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

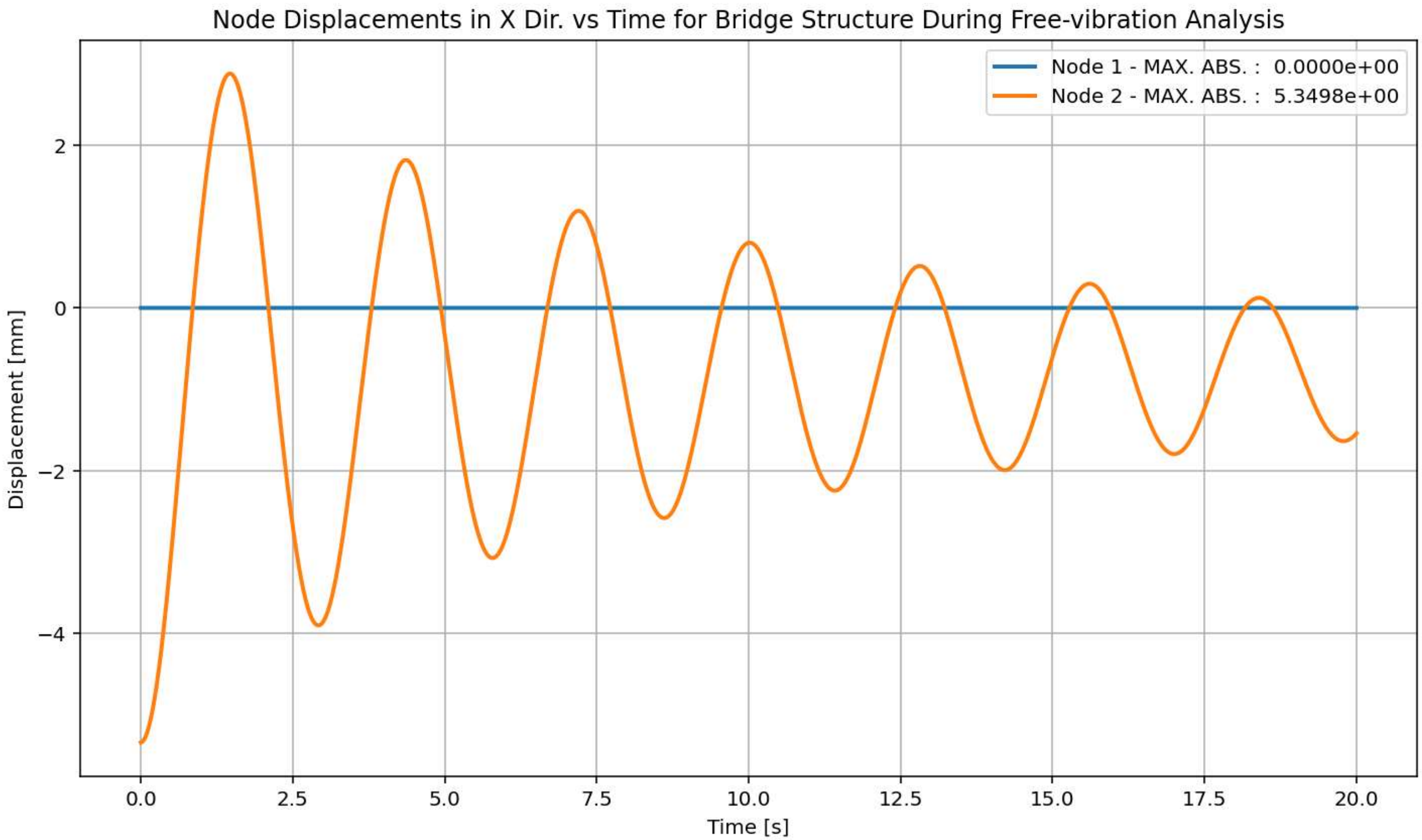


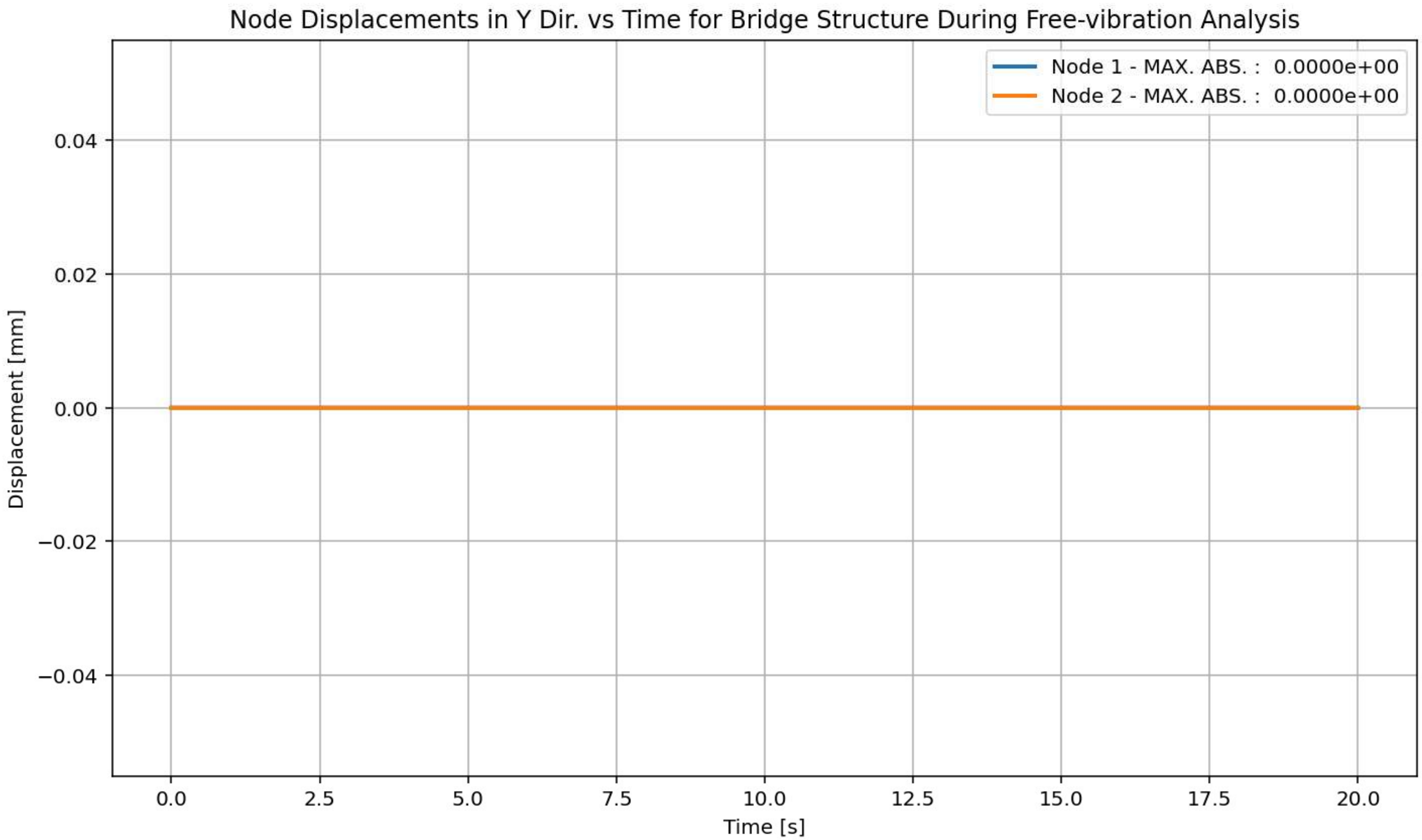
elements Stress in X Dir. vs Time for Truss Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 2.4021e+02
Element Stress [N/mm^2]
Time [s]
200
100
0
−100
−200
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



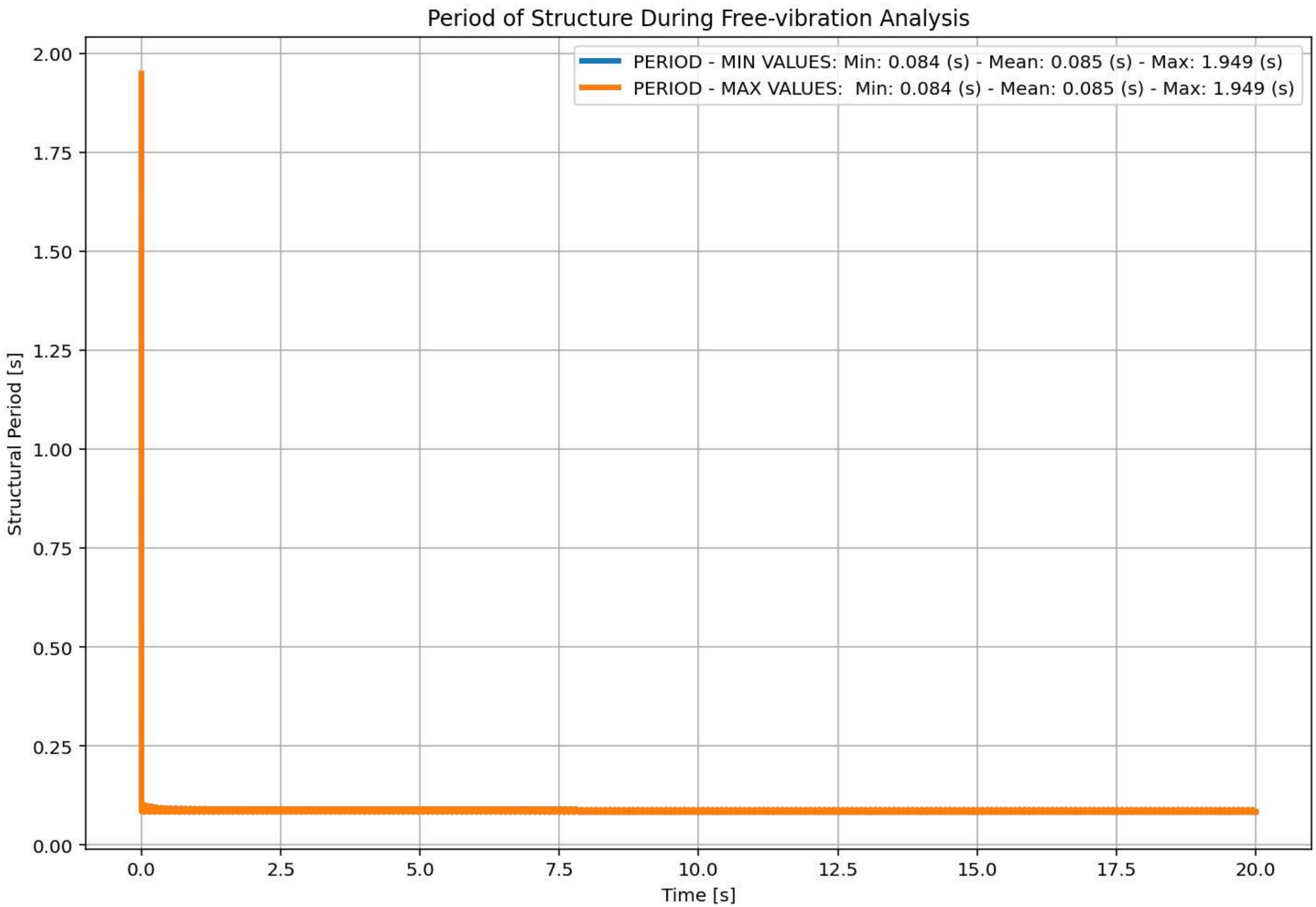
elements Strain in X Dir. vs Time for Truss Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 1.7833e-03
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
Element Strain [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


elements Damage-index [%] in X Dir. vs Time for Truss Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 1.7833e-03
Element Damage-index [%]
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

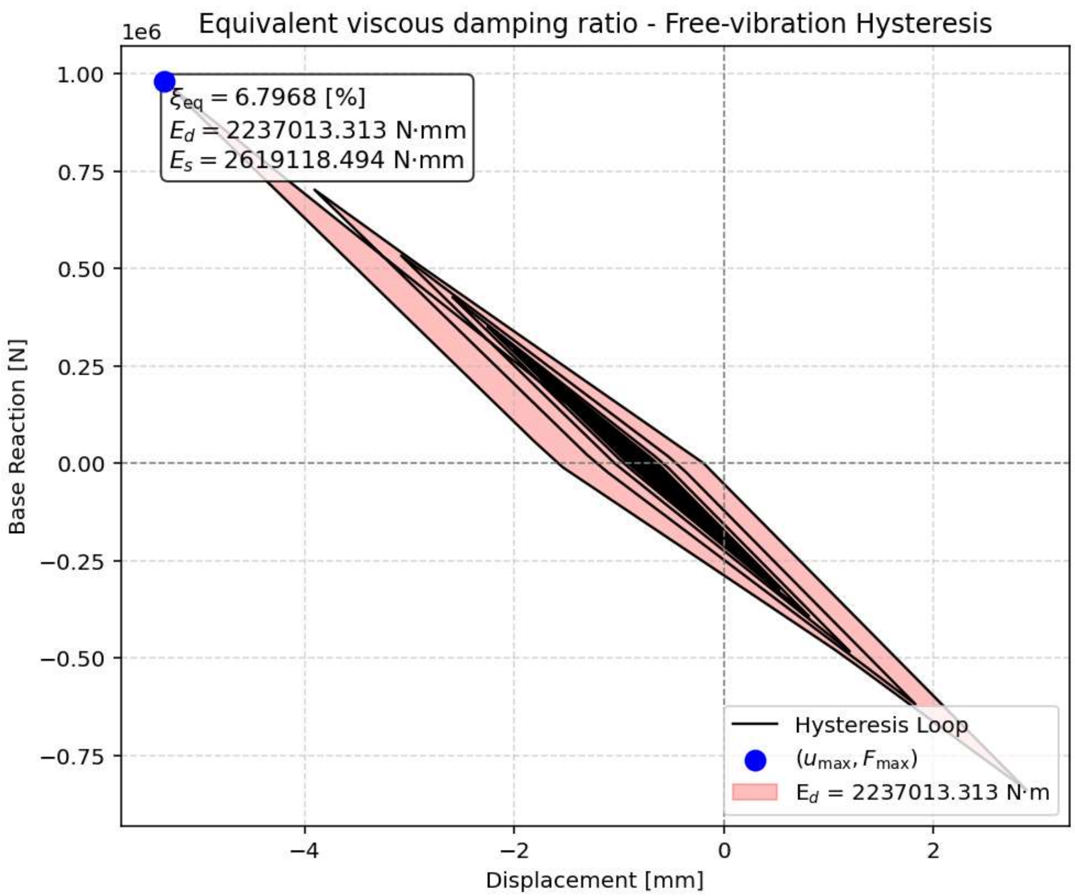

Node Displacements in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.3498e+00
Displacement [mm]
Time [s]
2
0
−2
−4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

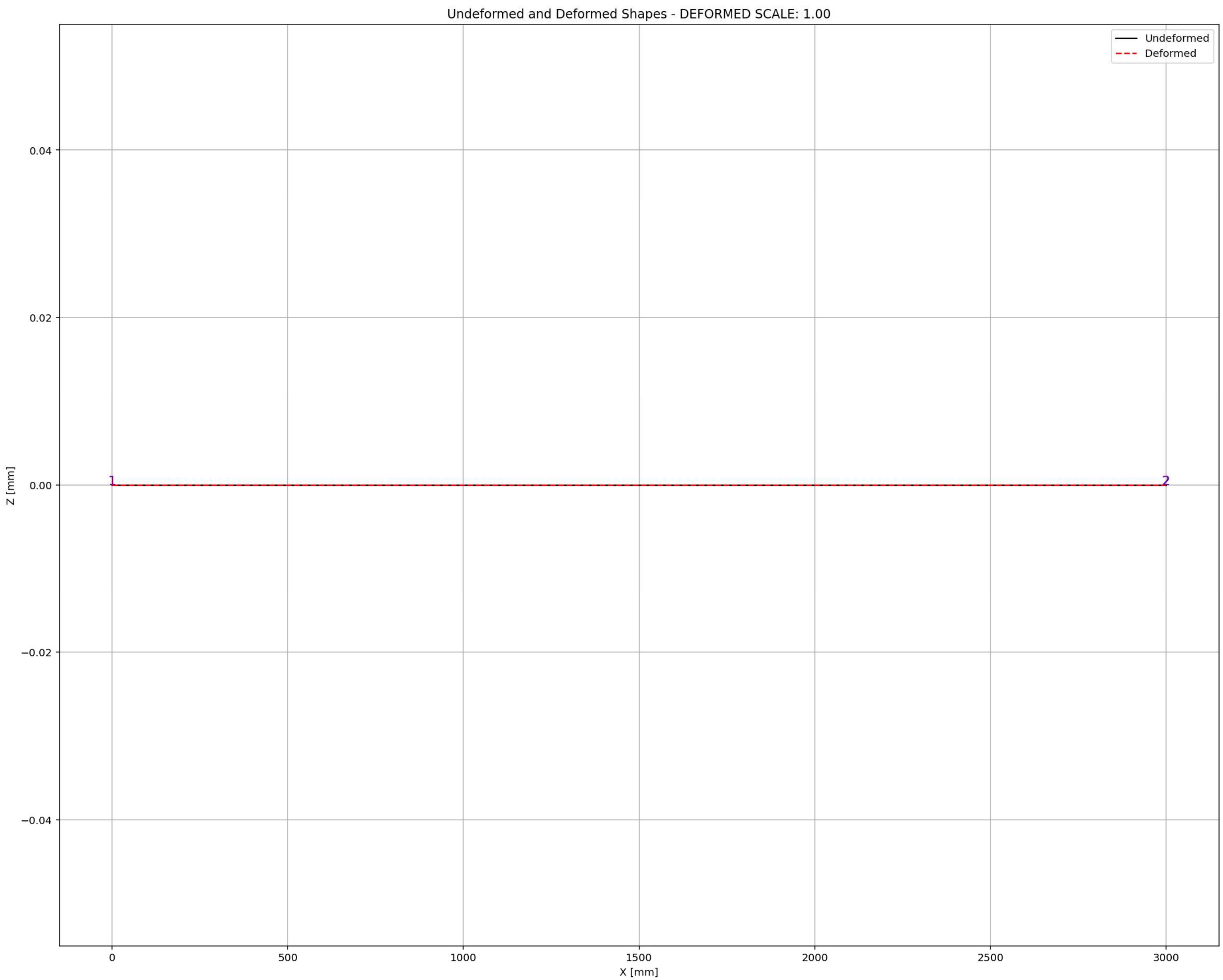


Node Displacements in Y Dir. vs Time for Bridge Structure During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 0.0000e+00
0.04
0.02
0.00
−0.02
−0.04
Displacement [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

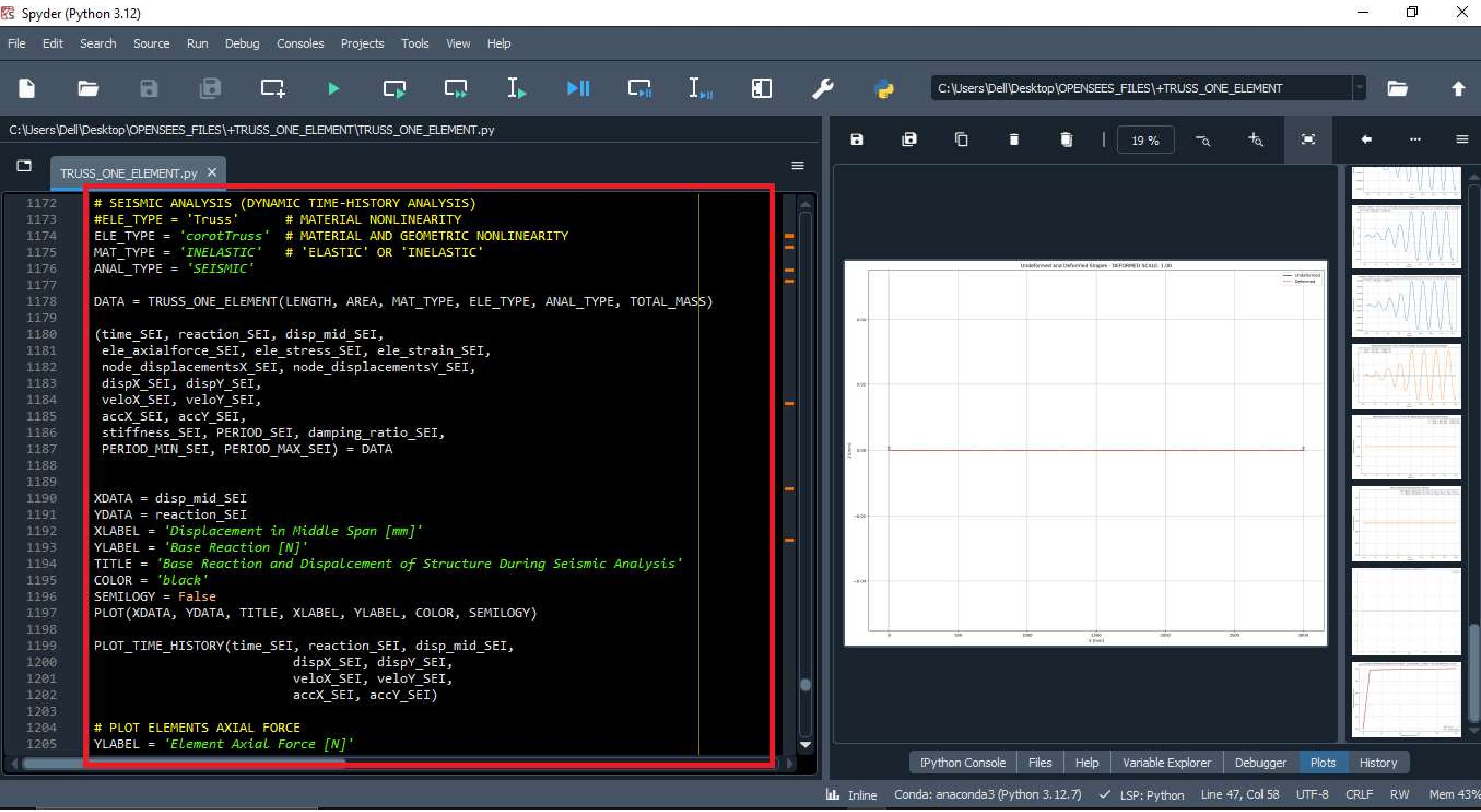


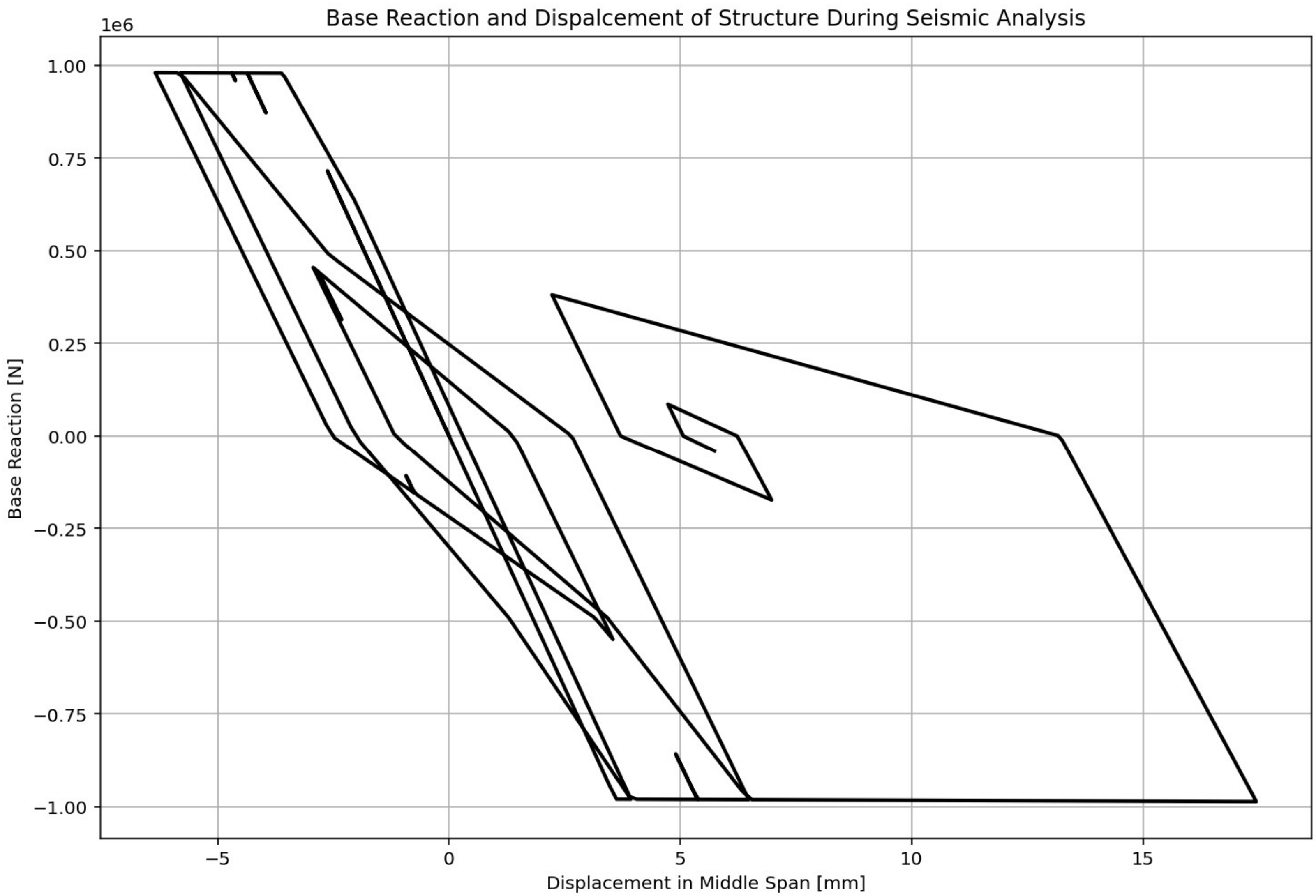
Period of Structure During Free-vibration Analysis
PERIOD - MIN VALUES: Min: 0.084 (s) - Mean: 0.085 (s) - Max: 1.949 (s)
PERIOD - MAX VALUES: Min: 0.084 (s) - Mean: 0.085 (s) - Max: 1.949 (s)
Structural Period [s]
Time [s]
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

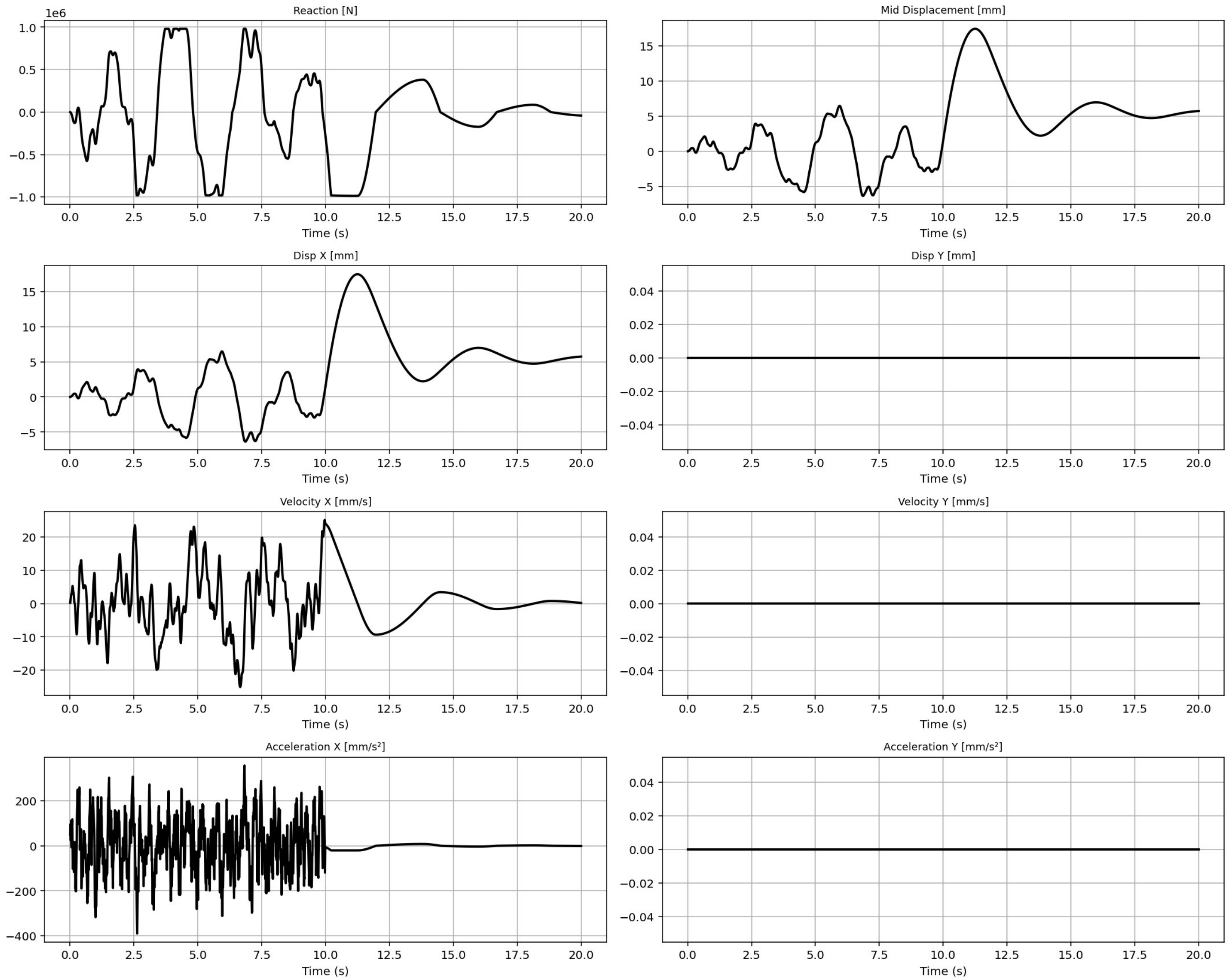

Equivalent viscous damping ratio - Free-vibration Hysteresis
1e6
$\xi_{eq} = 6.7968$ [%]
$E_d = 2237013.313$ N·mm
$E_s = 2619118.494$ N·mm
Base Reaction [N]
Displacement [mm]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−4
−2
0
2
Hysteresis Loop
(u_{max}, F_{max})
E_d = 2237013.313 N·m

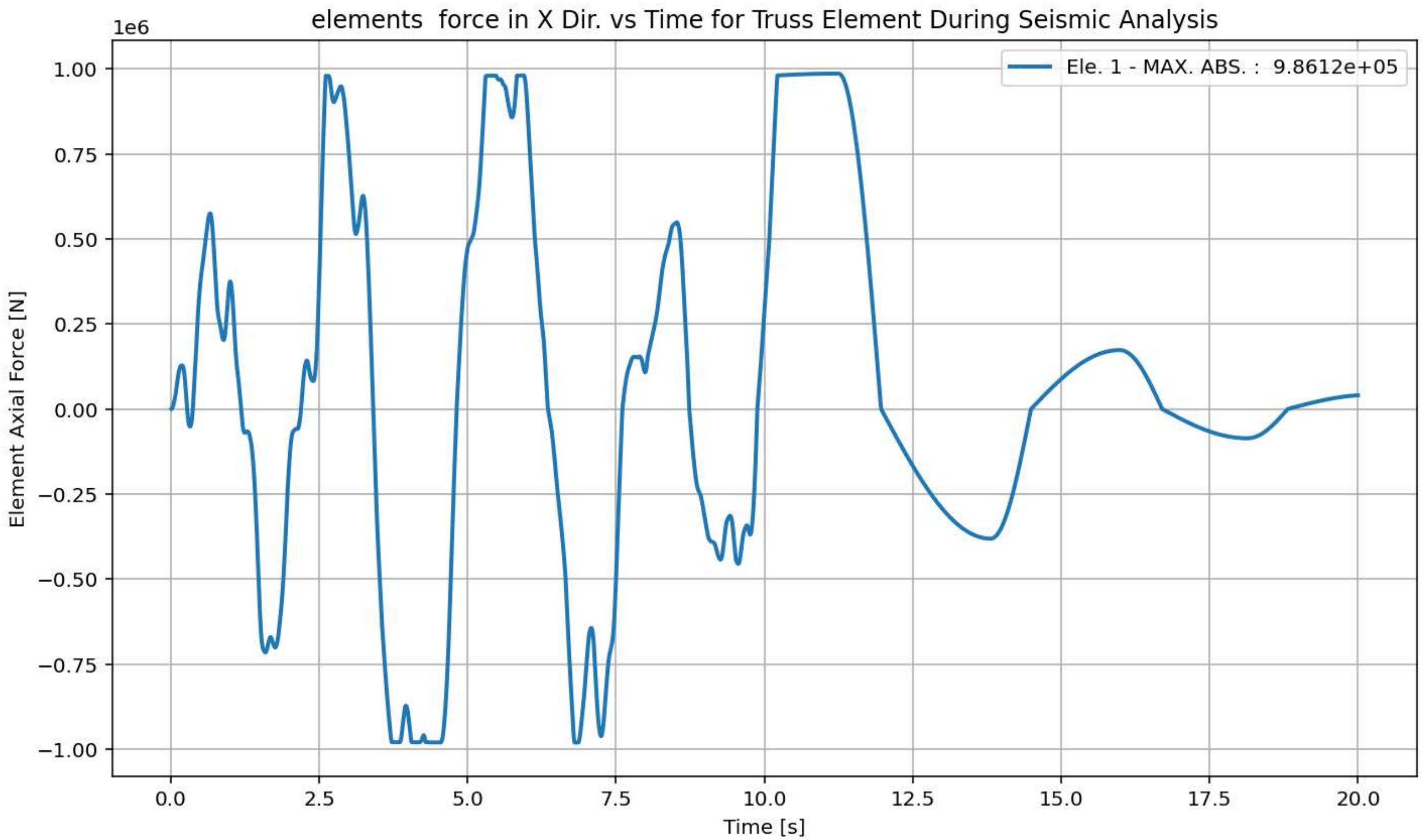

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.04
0.02
0.00
−0.02
−0.04
0
500
1000
1500
2000
2500
3000
1
2

SEISMIC ANALYSIS

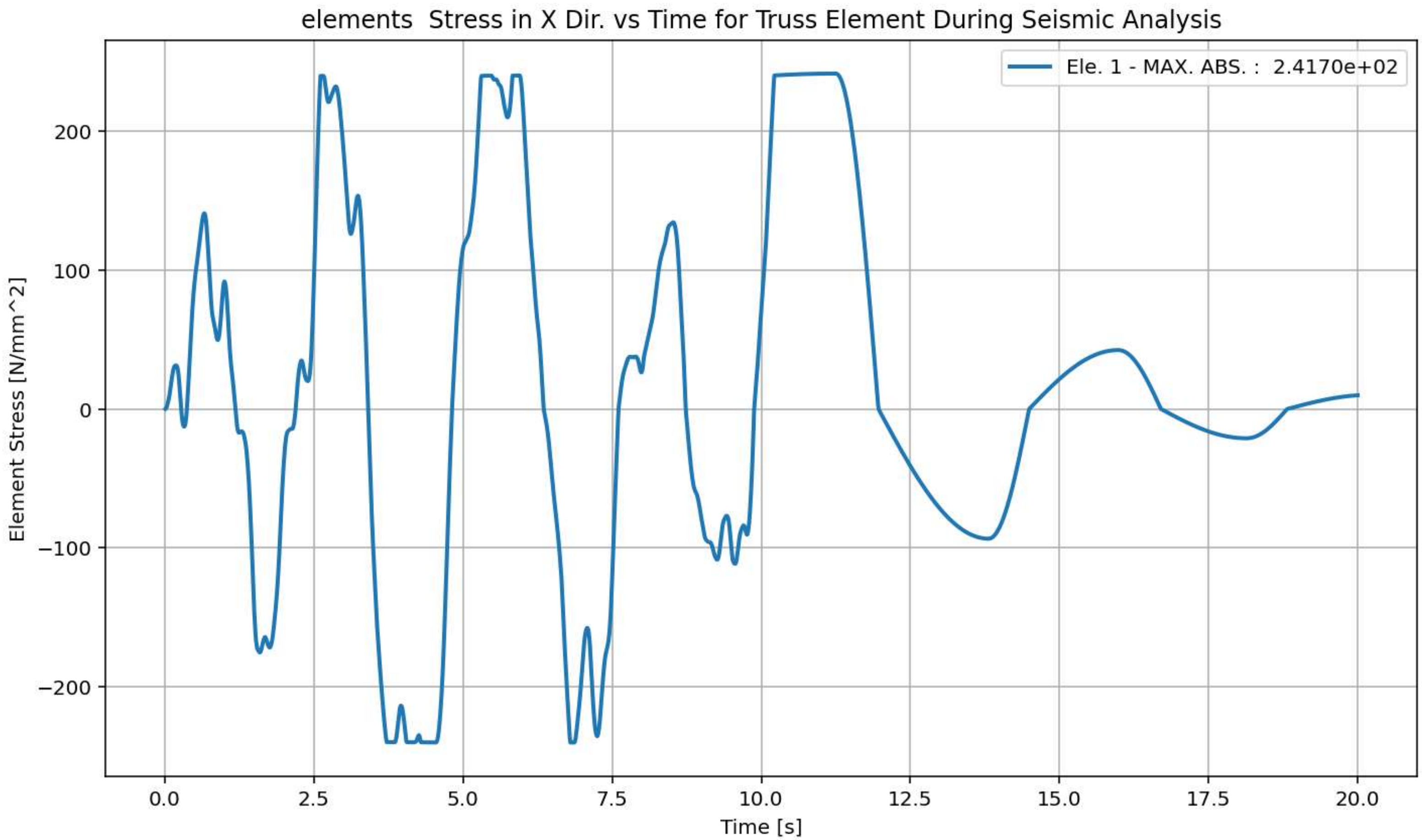

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT
C:\Users\Dell\Desktop\OPENSEES_FILES\+TRUSS_ONE_ELEMENT\TRUSS_ONE_ELEMENT.py
TRUSS_ONE_ELEMENT.py
SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITY
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'SEISMIC'
DATA = TRUSS_ONE_ELEMENT(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(time_SEI, reaction_SEI, disp_mid_SEI,
 ele_axialforce_SEI, ele_stress_SEI, ele_strain_SEI,
 node_displacementsX_SEI, node_displacementsY_SEI,
 dispX_SEI, dispY_SEI,
 veloX_SEI, veloY_SEI,
 accX_SEI, accY_SEI,
 stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
 PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
XDATA = disp_mid_SEI
YDATA = reaction_SEI
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_mid_SEI,
                  dispX_SEI, dispY_SEI,
                  veloX_SEI, veloY_SEI,
                  accX_SEI, accY_SEI)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 47, Col 58 UTF-8 CRLF RW Mem 43%


Base Reaction and Dispalcement of Structure During Seismic Analysis
1e6
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Base Reaction [N]
−5
0
5
10
15
Displacement in Middle Span [mm]

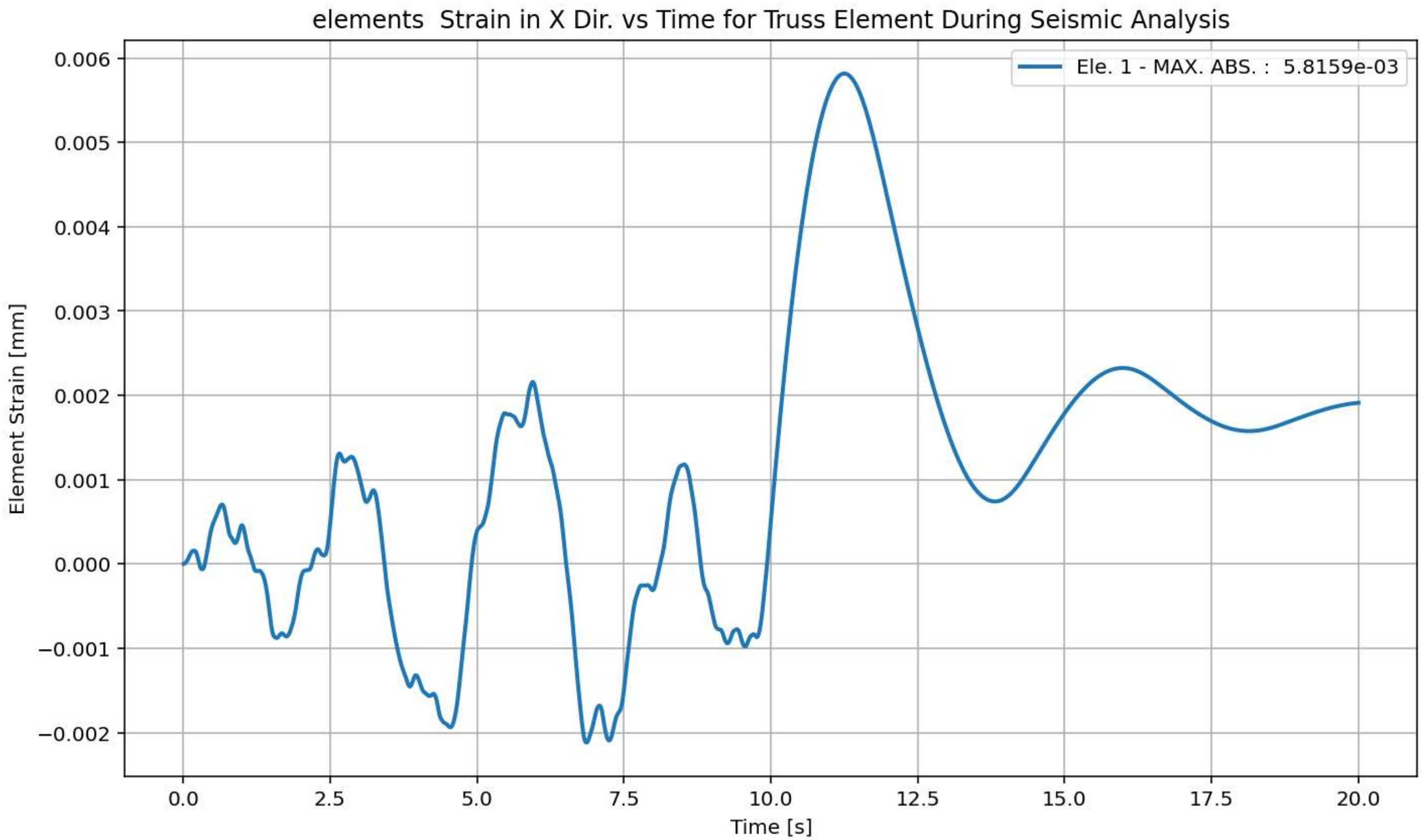

Reaction [N]
1e6
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

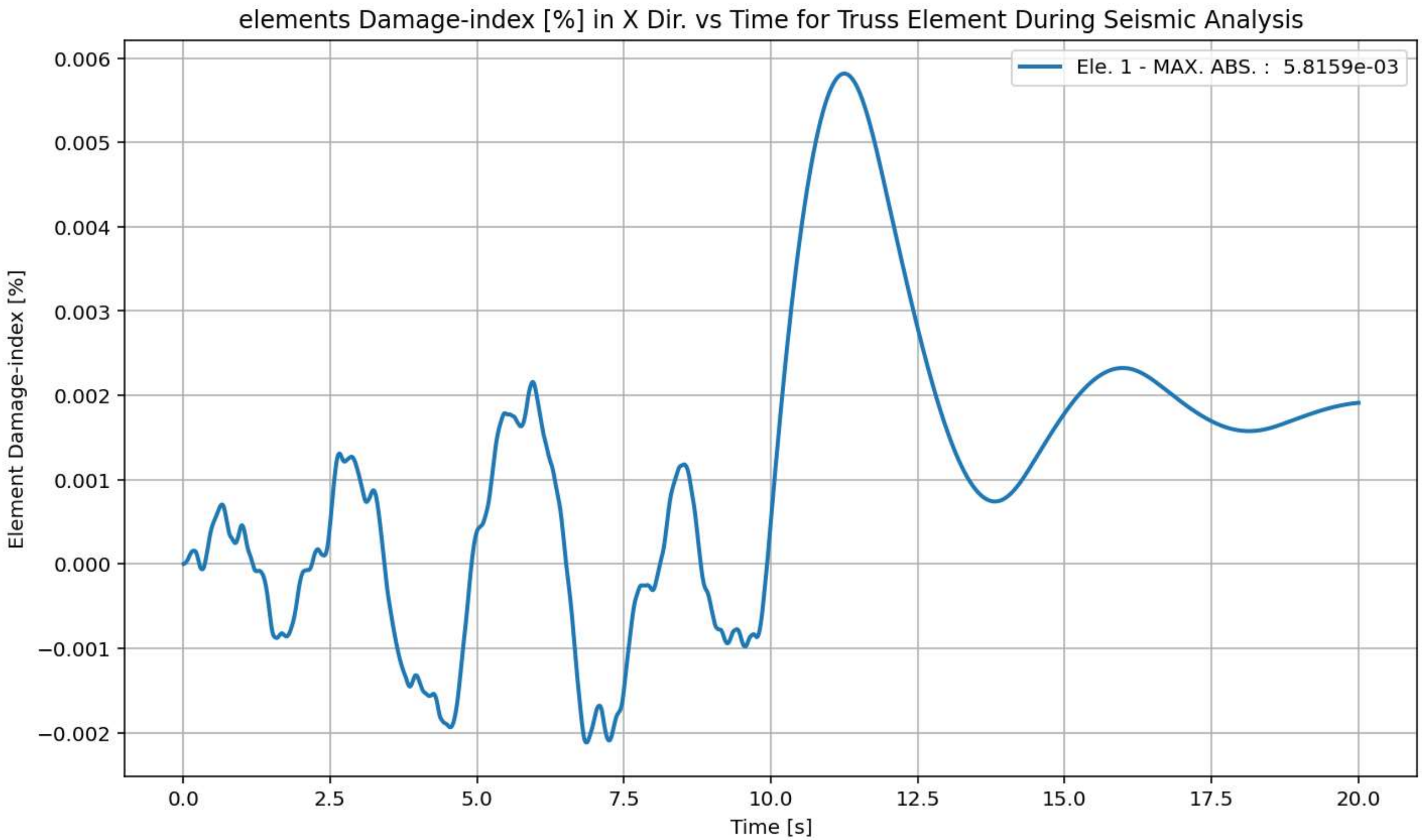


elements force in X Dir. vs Time for Truss Element During Seismic Analysis
1e6
Ele. 1 - MAX. ABS. : 9.8612e+05
Element Axial Force [N]
Time [s]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

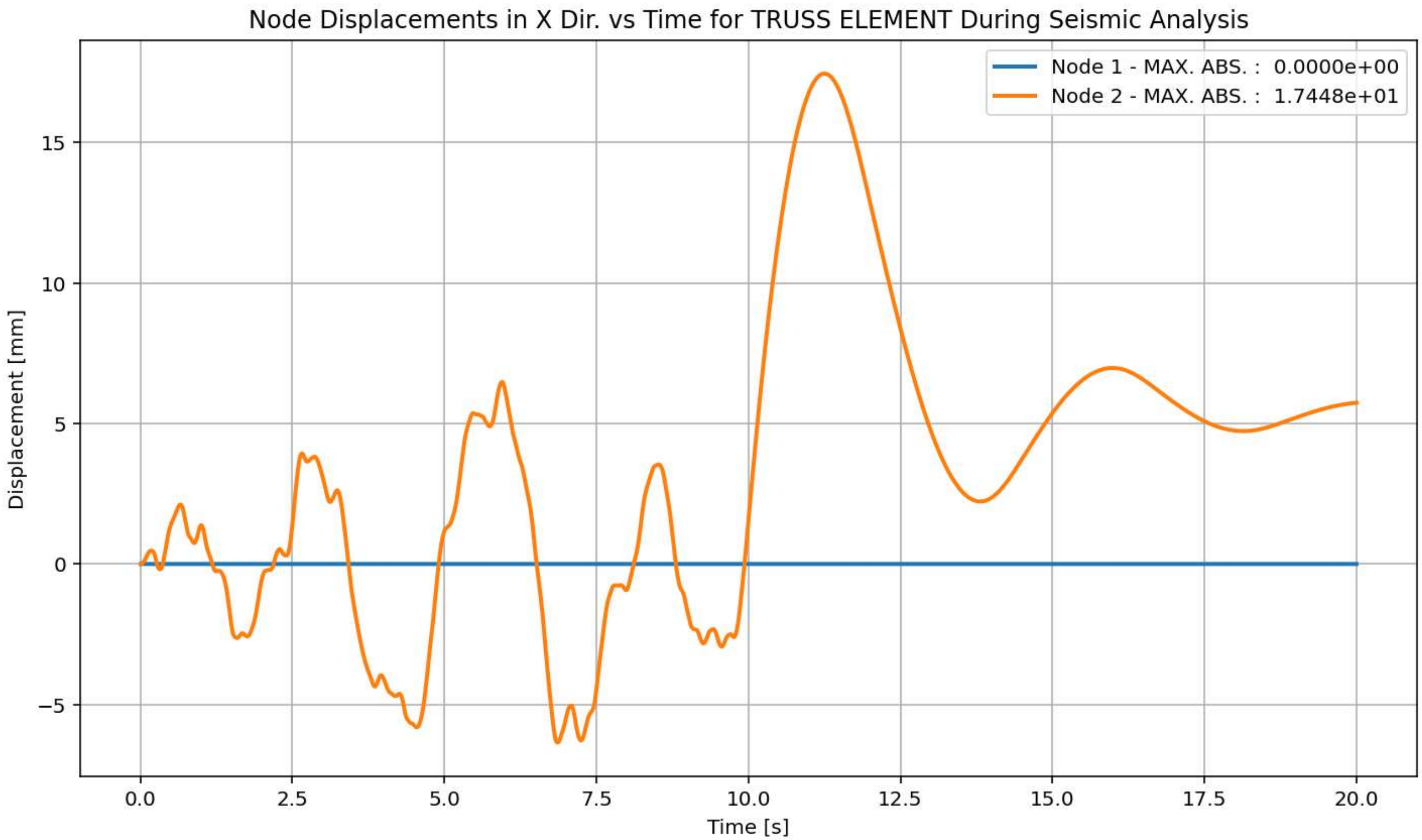


elements Stress in X Dir. vs Time for Truss Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 2.4170e+02
Element Stress [N/mm^2]
200
100
0
−100
−200
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

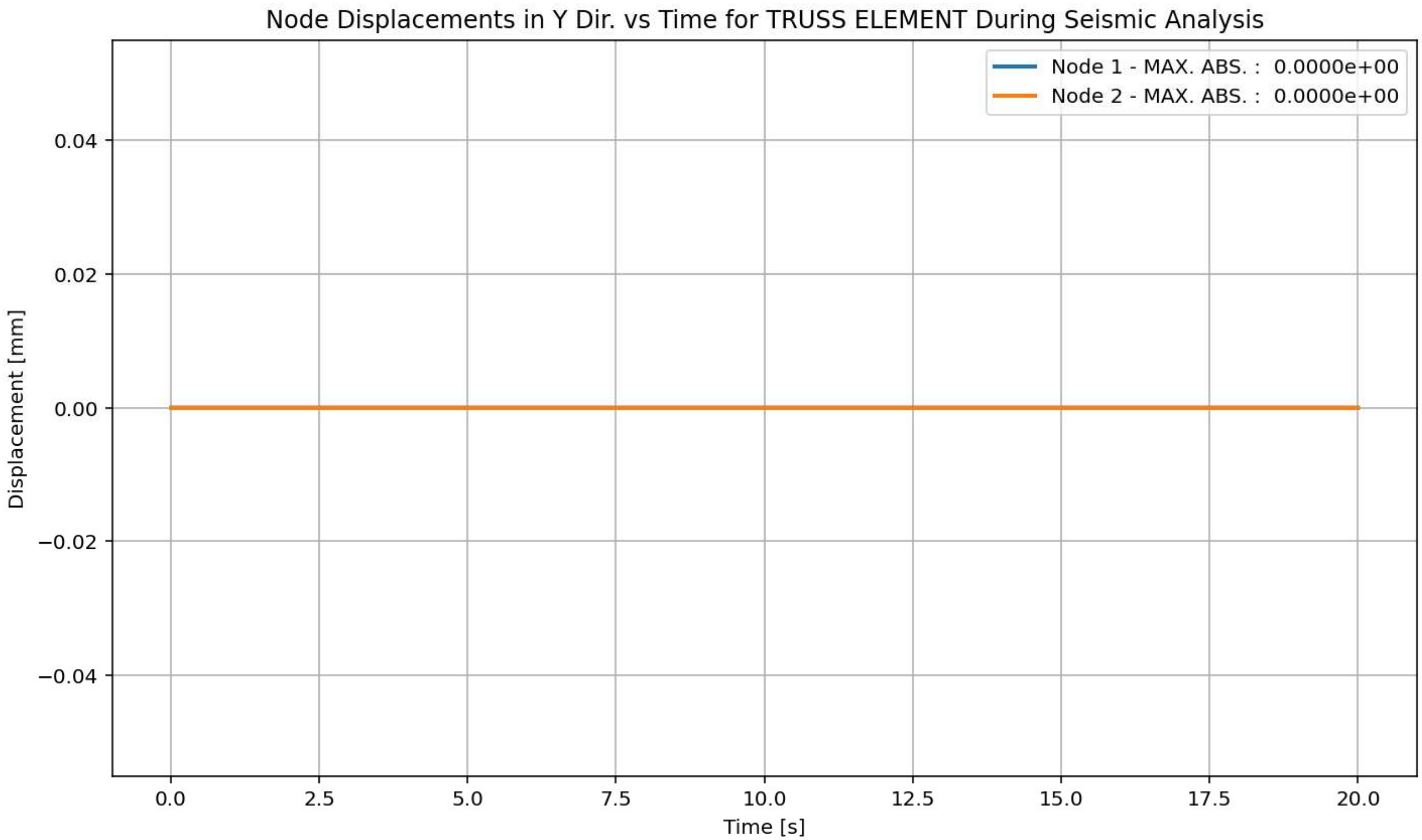

elements Strain in X Dir. vs Time for Truss Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 5.8159e-03
Element Strain [mm]
Time [s]
0.006
0.005
0.004
0.003
0.002
0.001
0.000
−0.001
−0.002
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



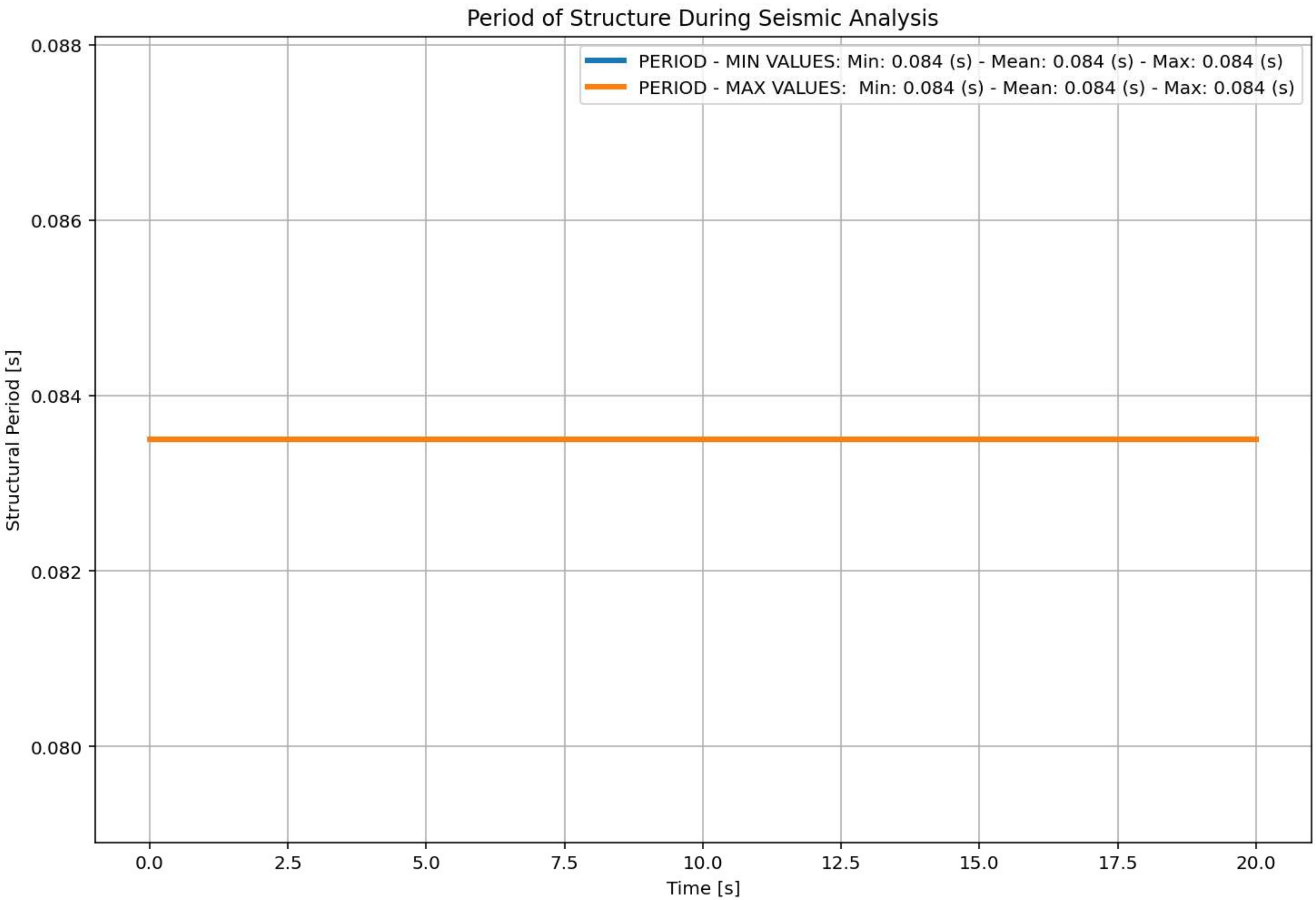
elements Damage-index [%] in X Dir. vs Time for Truss Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 5.8159e-03
Element Damage-index [%]
Time [s]
0.006
0.005
0.004
0.003
0.002
0.001
0.000
−0.001
−0.002
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



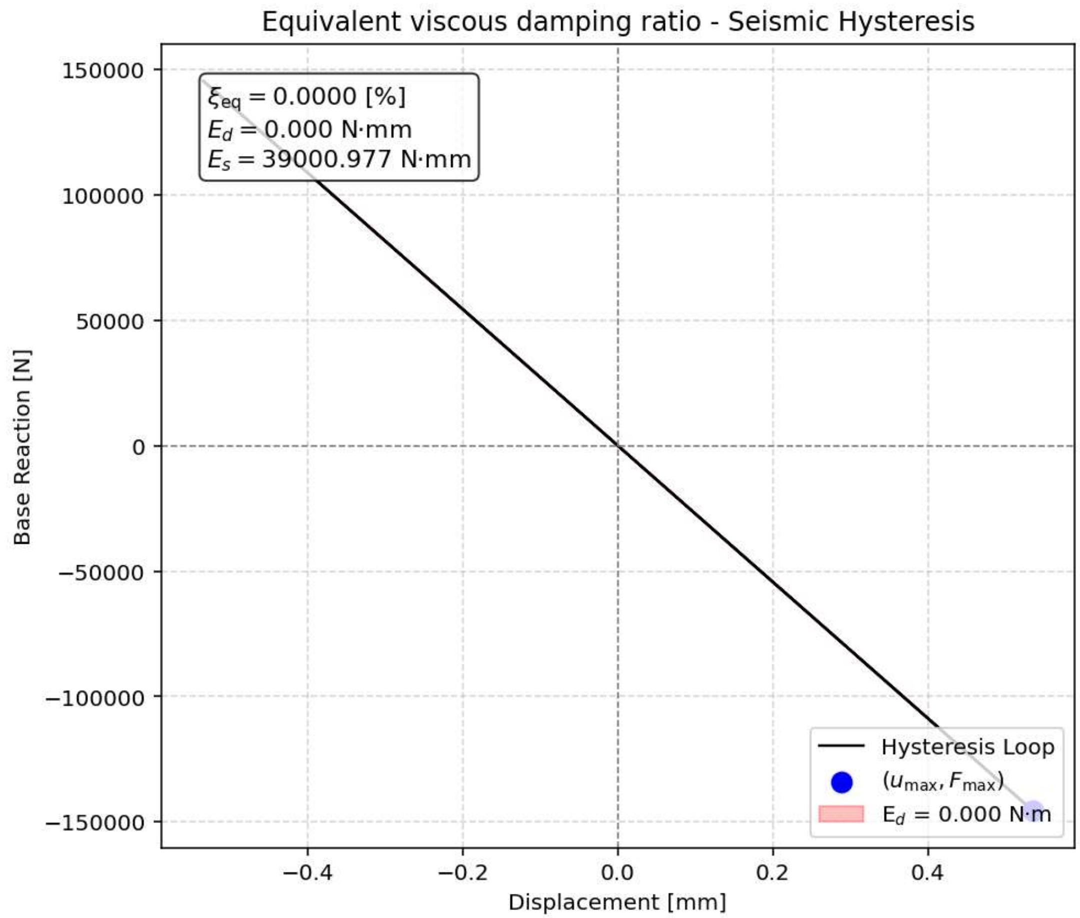
Node Displacements in X Dir. vs Time for TRUSS ELEMENT During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.7448e+01
Displacement [mm]
Time [s]
−5
0
5
10
15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

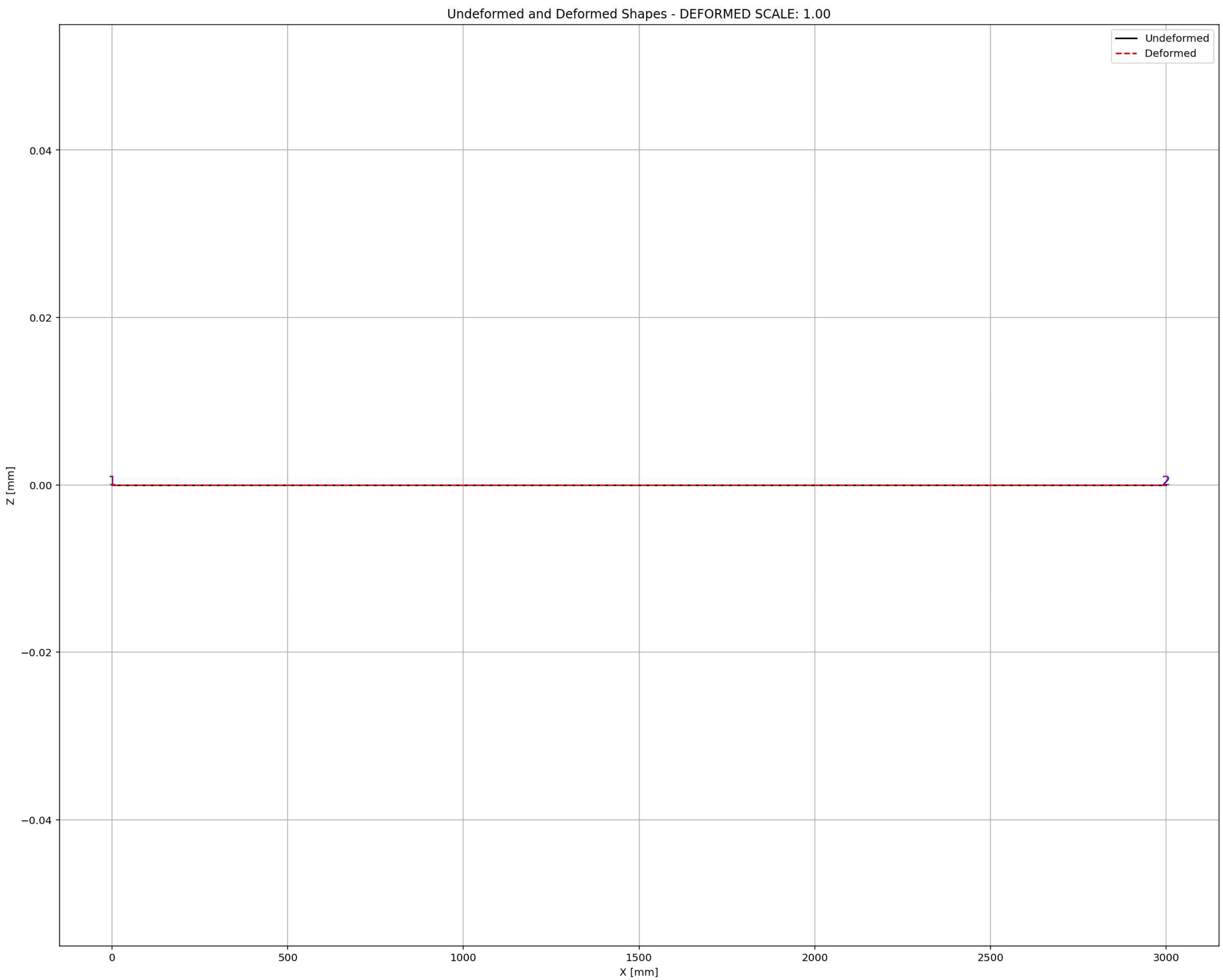


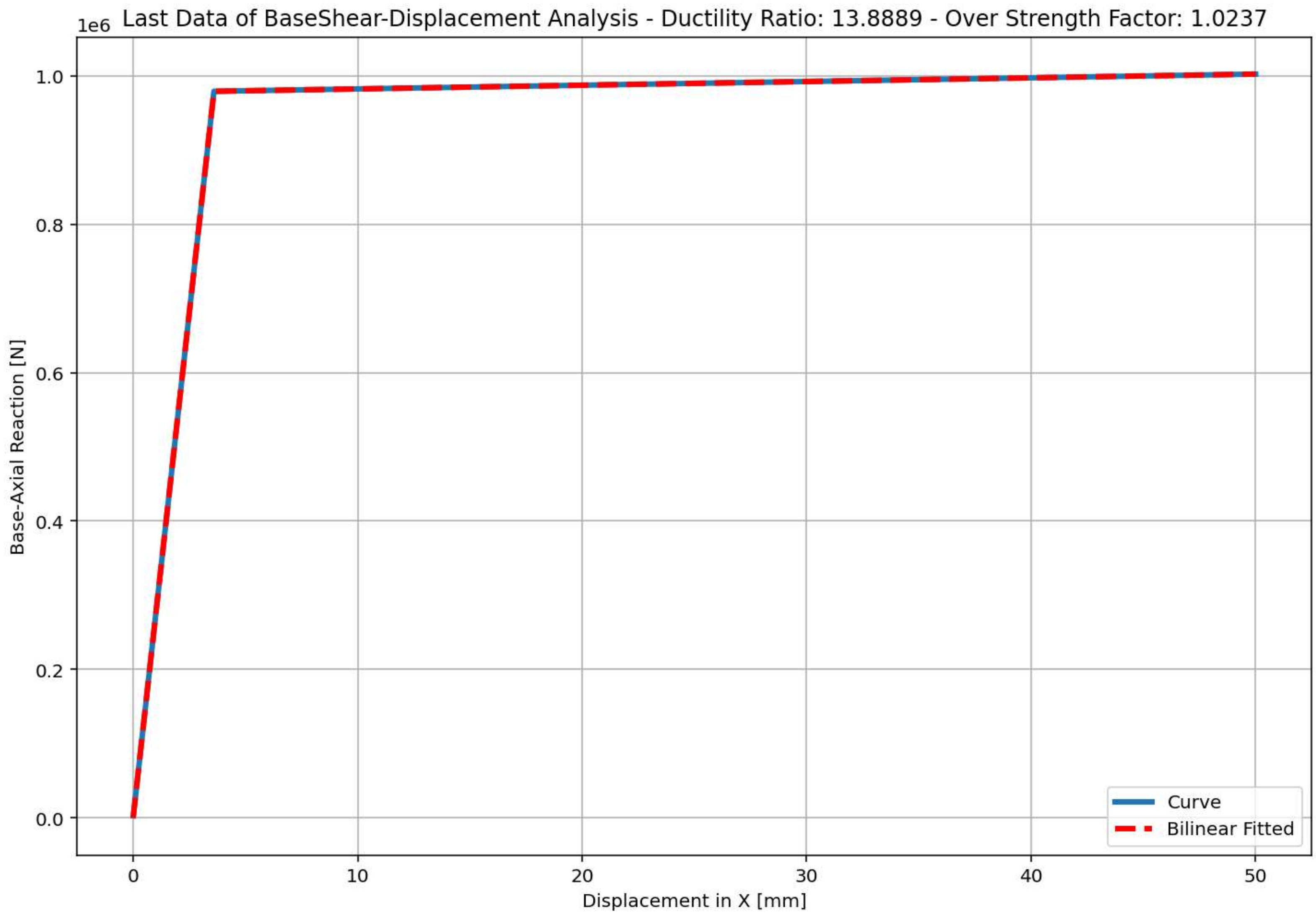
Node Displacements in Y Dir. vs Time for TRUSS ELEMENT During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 0.0000e+00
0.04
0.02
0.00
−0.02
−0.04
Displacement [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]



Period of Structure During Seismic Analysis
PERIOD - MIN VALUES: Min: 0.084 (s) - Mean: 0.084 (s) - Max: 0.084 (s)
PERIOD - MAX VALUES: Min: 0.084 (s) - Mean: 0.084 (s) - Max: 0.084 (s)
0.088
0.086
0.084
0.082
0.080
Structural Period [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

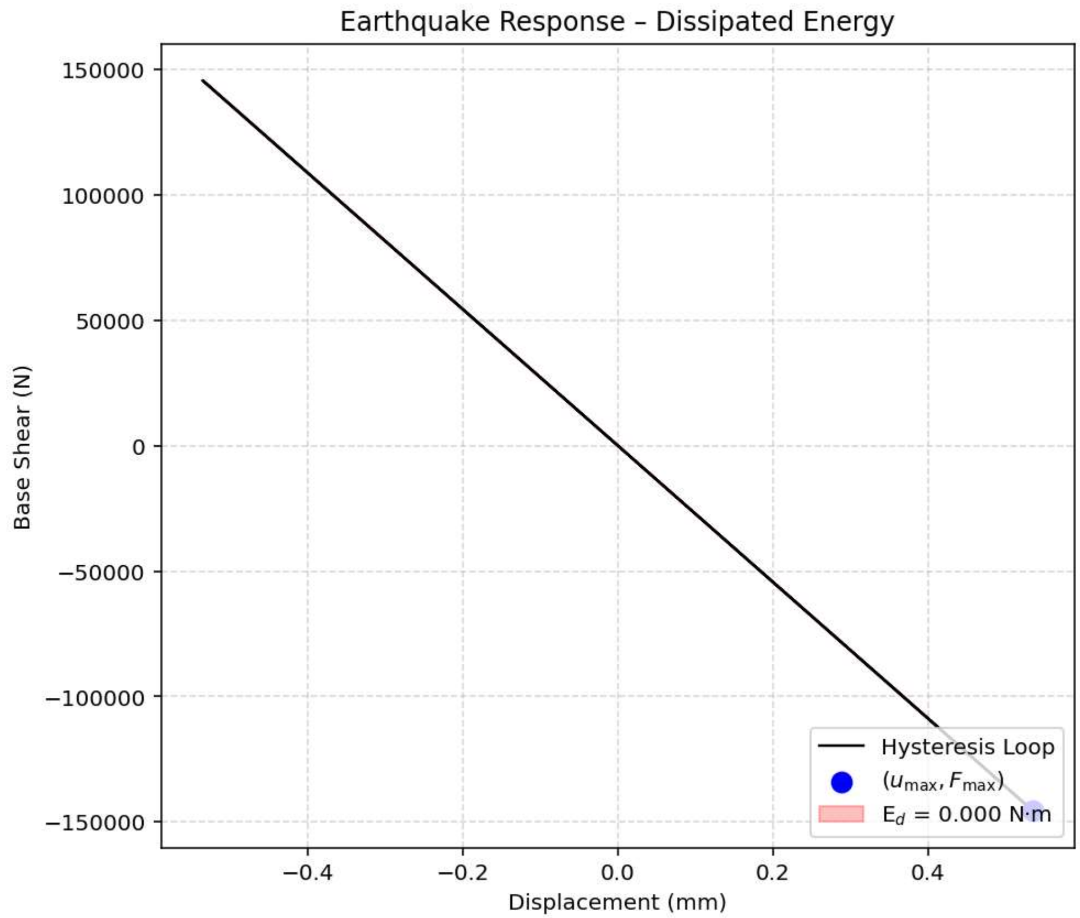

Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 0.0000$ [%]
$E_d = 0.000$ N·mm
$E_s = 39000.977$ N·mm
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.000 N·m
Base Reaction [N]
Displacement [mm]
150000
100000
50000
0
−50000
−100000
−150000
−0.4
−0.2
0.0
0.2
0.4


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0.04
0.02
0.00
−0.02
−0.04
0
500
1000
1500
2000
2500
3000
1
2

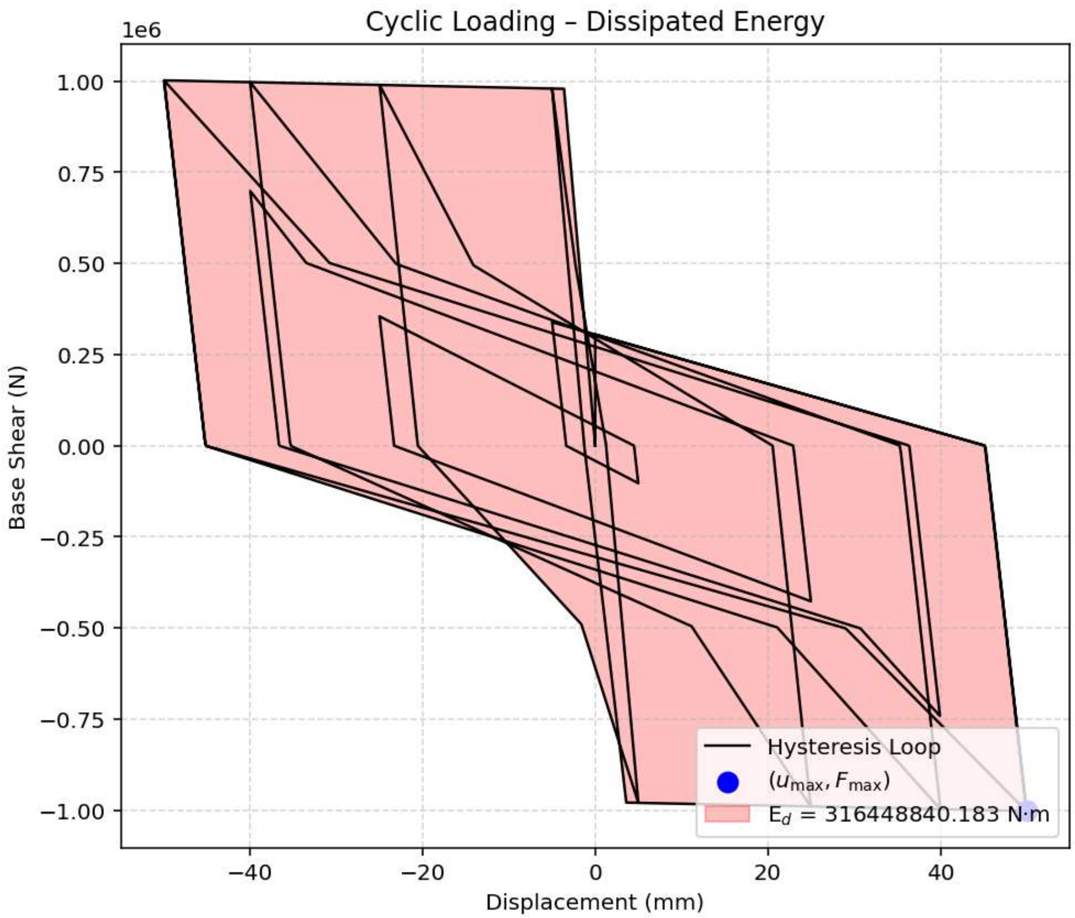


Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 13.8889 - Over Strength Factor: 1.0237
1e6
1.0
0.8
0.6
0.4
0.2
0.0
Base-Axial Reaction [N]
0
10
20
30
40
50
Displacement in X [mm]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
----------------------------
[[0.00000000e+00 0.00000000e+00]
 [3.59999784e+00 9.79199413e+05]
 [5.00000000e+01 1.00237638e+06]]
+==========================+
+--------------------------------------------------+
 Structure Elastic Stiffness :     272000.00
 Structure Plastic Stiffness :     20047.53
 Structure Tangent Stiffness :     499.50
 Structure Ductility Ratio :      13.89
 Structure Over Strength Factor:   1.02
+--------------------------------------------------+
Over Strength Coefficient (Ω0):      1.0237
Displacement Ductility Ratio (μ):    13.8889
Ductility Coefficient (Rμ):          13.8889
Structural Behavior Coefficient (R): 14.2176
Structural Ductility Damage Index in X Direction: 29.8445 (%)
```



Earthquake Response – Dissipated Energy
Base Shear (N)
150000
100000
50000
0
−50000
−100000
−150000
Displacement (mm)
−0.4
−0.2
0.0
0.2
0.4
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 0.000 N·m



Cyclic Loading – Dissipated Energy
1e6
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Base Shear (N)
−40
−20
0
20
40
Displacement (mm)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 316448840.183 N·m

Dissipated Energy from Earthquake= 0.00 N·mm
Dissipated Energy from Cyclic Displacement= 316448840.18 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.000 [%]
ZONE 1: VERY MINOR DAMAGE